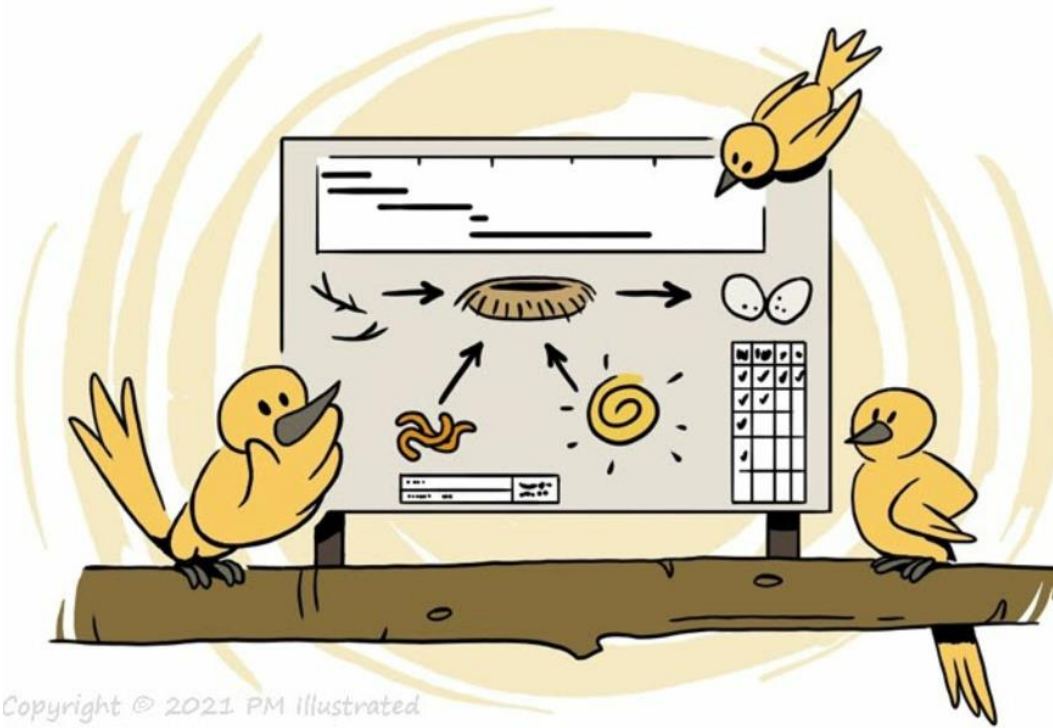


Project Management

6. Schedule Management

Contents

- Schedule Management
 - What is project schedule?
- Plan Schedule Management
- Define Activities
 - What is project activity?
 - What is milestone?
- Sequence Activities
- Estimate Activity Durations
- Develop Schedule
- Control Schedule



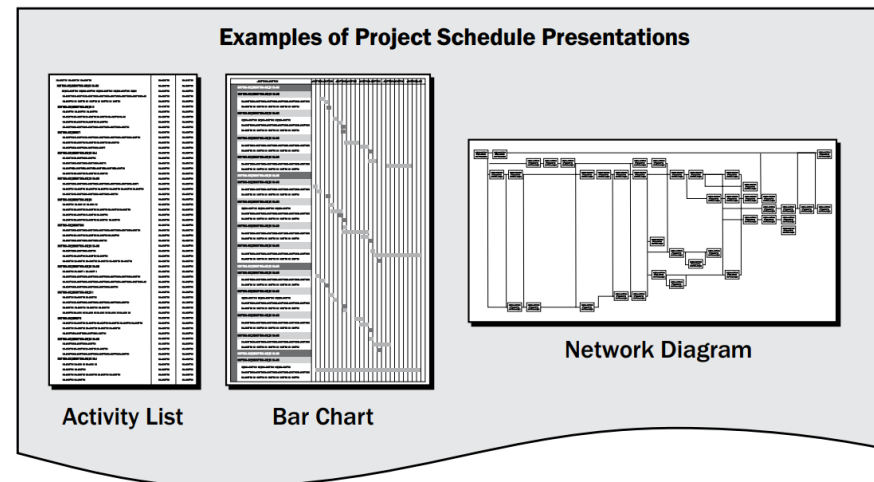
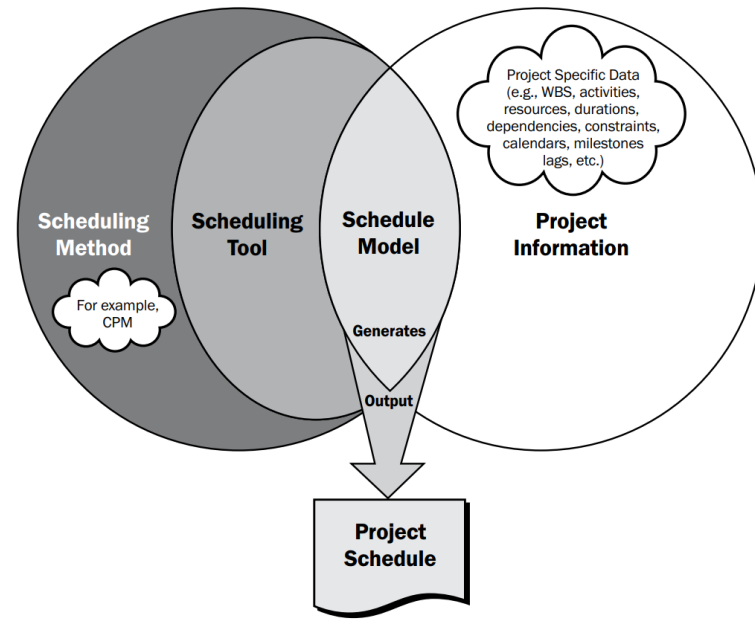
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Schedule Management

Project schedule management is what most people think of when they think of project managers.

What is project schedule?

- A project schedule is a model that presents linked activities with planned dates.
- Shows durations, milestones and the necessary people and resources required.
- When things start and end, and how activities roll up to milestones or releases.
- Once created, they can also be used to track actual project performance against and keep stakeholders informed of progress.



1

Plan Schedule Management



Schedule Management plan

First you define the processes you'll use to plan and control your schedule.

3

Sequence Activities

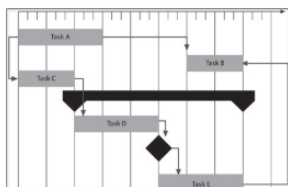


Project schedule network diagram

Next, you figure out which activities need to come before others, and put them in the right order. The main output here is a **project schedule network diagram**, a picture of how activities are related.

5

Develop Schedule



Then you build a schedule from all of the estimates and the resource and activity information you've created.

2

Define Activities



Activity list

Next you come up with a list of all of the activities that will need to be completed.

4

Estimate Activity Durations



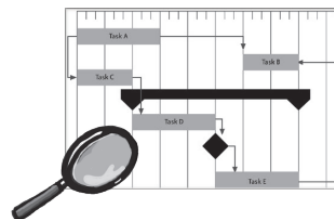
Activity duration estimates

...and then estimate the time it will take to do each activity.

You can use lots of different estimation techniques to determine how long the project will take.

6

Control Schedule

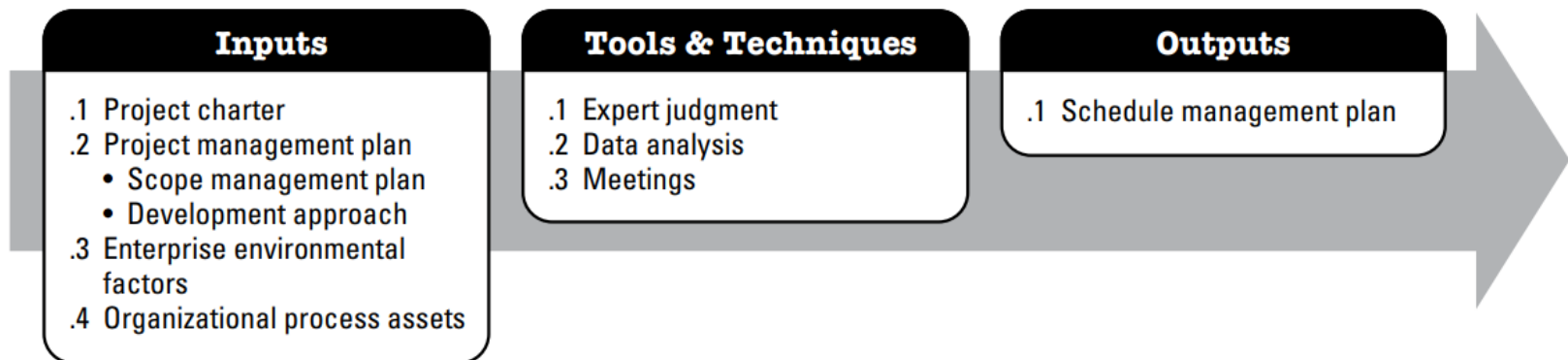


Finally, you monitor and control changes to the schedule to make sure that it is kept up to date.

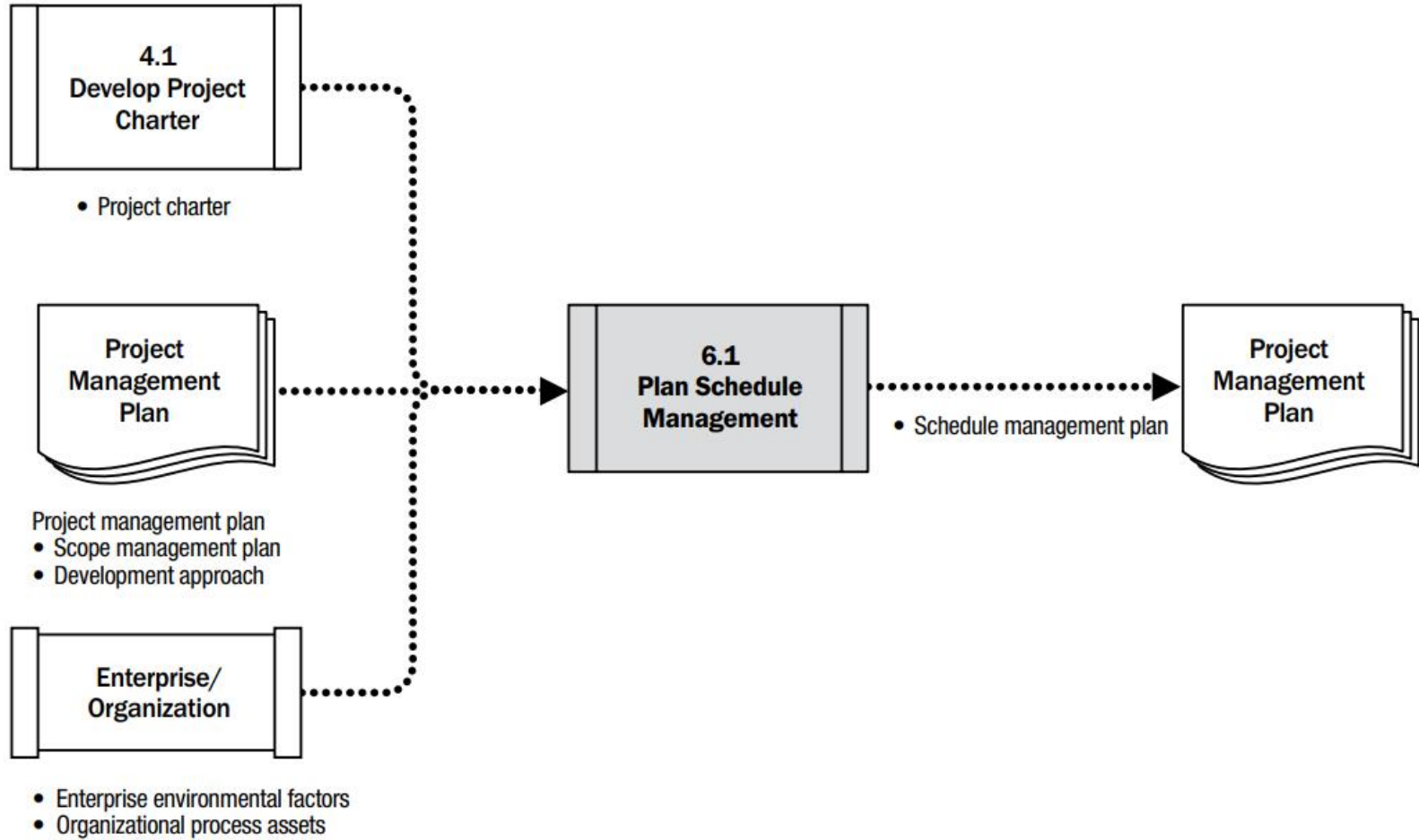
Plan Schedule Management

Plan Scope Management

- **What:** process of establishing the policies, procedures, and documentation for planning, developing, managing, executing, and controlling the project schedule.
- **Why:** provides guidance and direction on how the project schedule will be managed throughout the project.
- **When:** performed once or at predefined points in the project.



Data Flow



Outputs

Schedule management plan

- Formal or informal, highly detailed, or broadly framed based on the needs of the project
- That establishes the procedure, policies and the activities for developing, monitoring, and controlling the schedule.
- A component of the project management plan

SCHEDULE MANAGEMENT PLAN		
Project Title: _____ Date: _____		
Schedule Methodology: 		
Scheduling Tools: 		
Level of Accuracy:	Units of Measure:	Variance Thresholds:
Schedule Reporting and Format: 		
Organizational Procedure Links: 		
Schedule Updates: 		
Page 1 of 1		

Define Activities

Define Activities

- **What:** process of identifying and documenting the **specific actions** to be performed **to produce the project deliverables**.
- **Why:** decomposes work packages into schedule activities.
- **When:** performed once or at predefined points in the project.
- **How:**
 - Start with the Work Breakdown Structure (WBS), which breaks the project scope into manageable work packages.
 - Decompose these work packages into smaller, actionable activities that can be assigned, scheduled, and tracked.
 - Example: For a work package like "Develop Website," activities might include "Design Homepage," "Code Backend," and "Test Functionality."



Work Package = What to Deliver

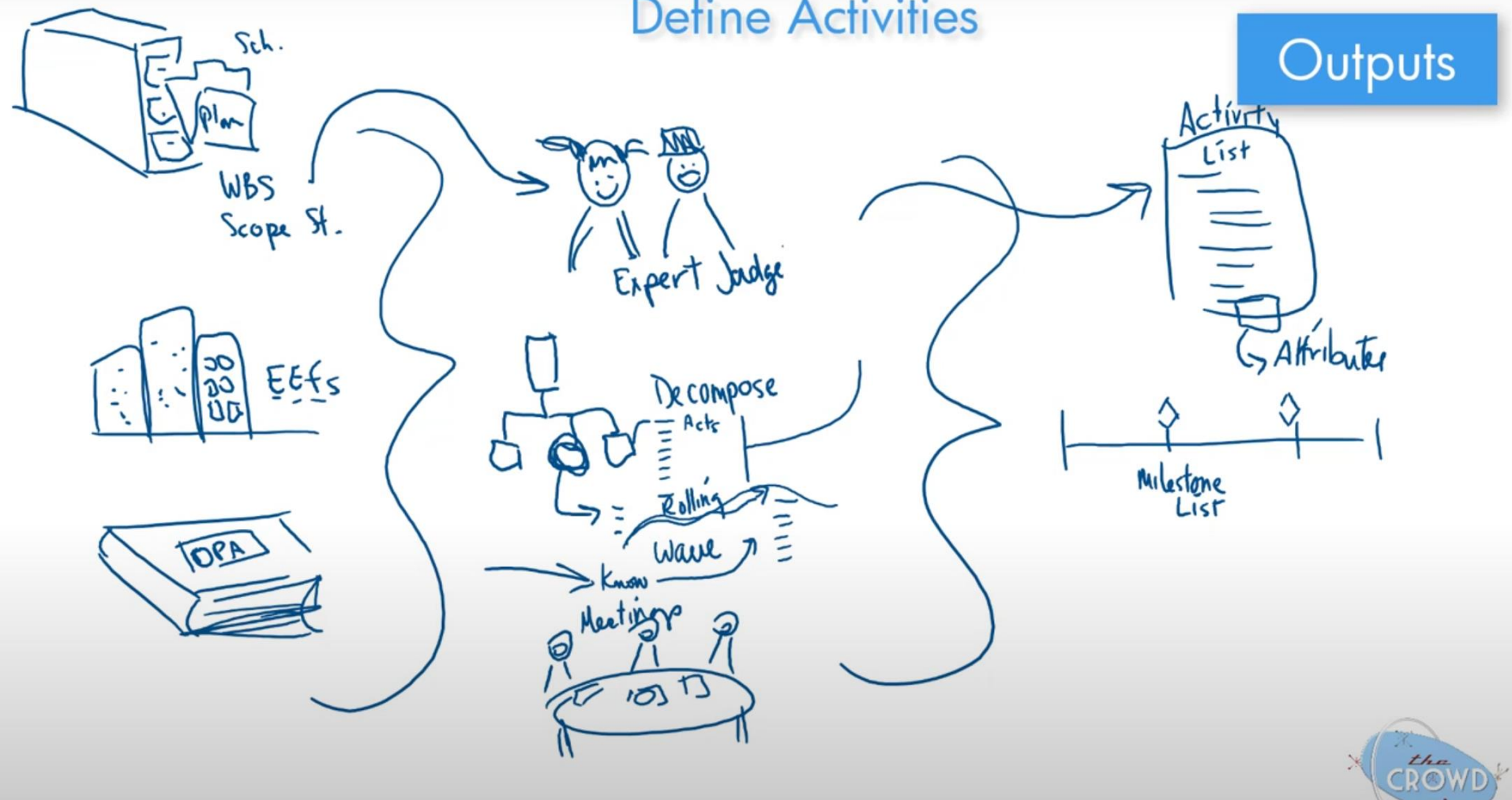
- Focus: **Scope**
- Describes: A **specific deliverable or output**
- Appears in: Work Breakdown Structure (WBS)
- Includes: Budget, responsible person/team, deliverable acceptance criteria

Activity = How and When to Do It

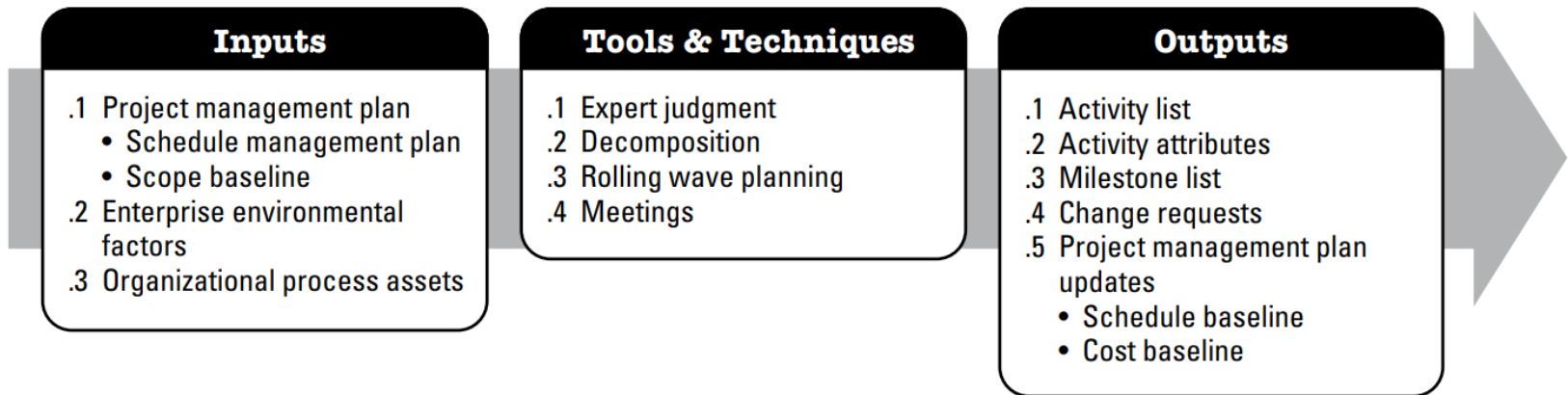
- Focus: **Time (Schedule)**
- Describes: The **steps needed to produce** the deliverable
- Appears in: **Project schedule (Gantt chart, network diagram)**
- Includes: Duration, start/end dates, dependencies, resources

Define Activities

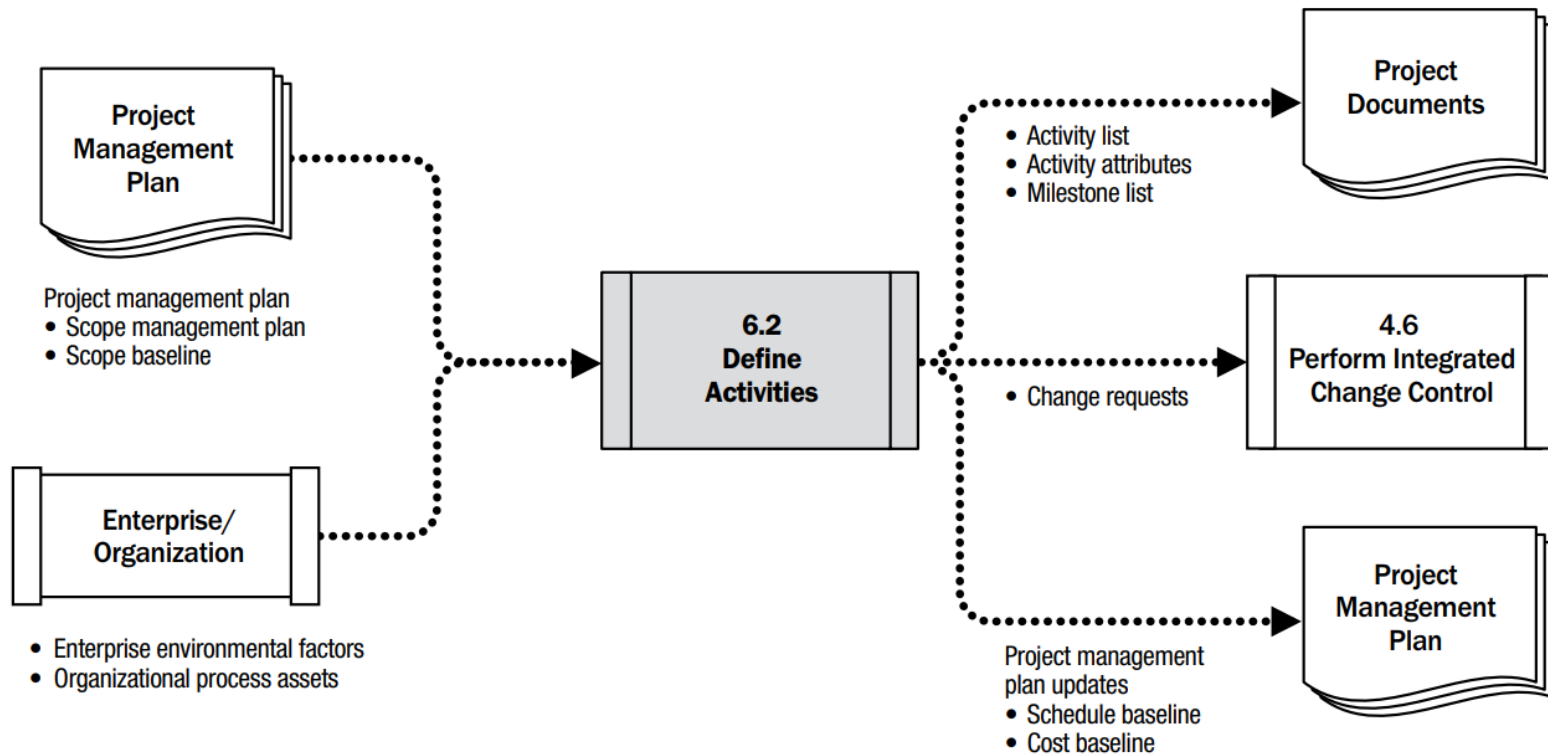
Outputs



Process



Data Flow



Outputs

Milestone List

- A **milestone list** is a project management document that identifies all project **milestones**.
- Type of milestone:
 - Internal or external (audience)
 - Interim or **final** (timing)
 - **Mandatory** or optional (importance)

MILESTONE LIST		
Project Title: _____		Date Prepared: _____
Milestone	Milestone Description	Type

Milestones

- **Marks completion** of major deliverables, key decisions, approvals, or phase transitions.
- A zero-duration event (unlike tasks, milestones don't consume time).
- Used to track progress, signal readiness to proceed, and facilitate stakeholder alignment.



Milestone	Date	Deliverables
Requirements Sign-off	June 10	Final approved SRS
Design Review Complete	June 25	UI/UX and System Architecture docs
MVP Deployment	August 5	First working version on staging
UAT Sign-off	August 20	User Acceptance Testing results
Final Release	September 1	Full deployment to production

Rule of Thumb:

- *Have at least one milestone per major phase or deliverable. For long projects, one per month or major decision point is a good pacing guideline.*
- *Fewer, well-defined milestones are better than many minor ones.*

Activity list

- [illegible]

Outputs

Activity attributes

- Describe the characteristics of the activities and are an extension of the activity list. The details may include but not limited:
 - predecessors,
 - successors,
 - logical relationships,
 - leads and lags,
 - resource requirements,
 - constraints,
 - imposed dates,
 - and assumptions related to the activity

ACTIVITY ATTRIBUTES					
Project Title: _____			Date Prepared: _____		
ID:	Activity:				
Description of Work:					
Predecessors	Relationship	Lead or Lag	Successor	Relationship	Lead or Lag
Number and Type of Team Resources Required:		Skill Requirements:		Required Resources:	
Type of Effort:					
Location of Performance:					
Imposed Dates or Other Constraints:					
Assumptions:					

Page 1 of 1



Work Package = What to Deliver

- Focus: **Scope**
- Describes: A **specific deliverable or output**
- Appears in: Work Breakdown Structure (WBS)
- Includes: Budget, responsible person/team, deliverable acceptance criteria

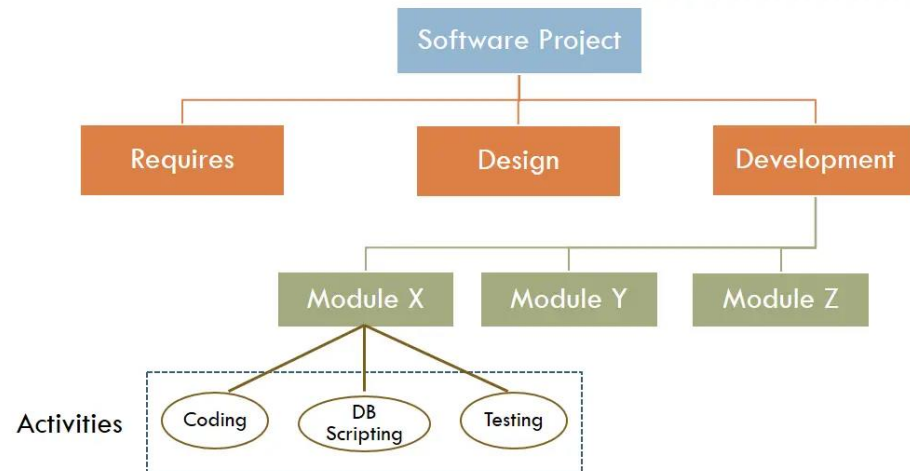
Activity = How and When to Do It

- Focus: **Time (Schedule)**
- Describes: The **steps needed to produce** the deliverable
- Appears in: **Project schedule (Gantt chart, network diagram)**
- Includes: Duration, start/end dates, dependencies, resources

Tools and Techniques

Decomposition

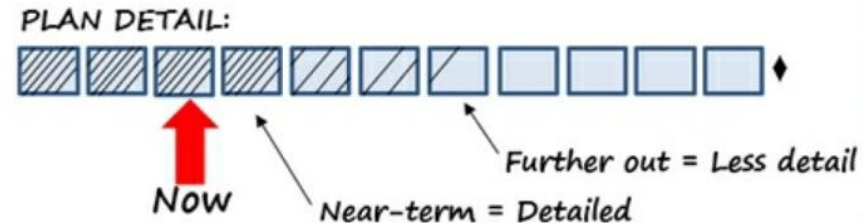
- Activities represent the effort needed to complete a work package.
- Decomposition is a technique used for dividing and subdividing the project scope and project deliverables into smaller, more manageable parts.
- When to stop decomposition?
 - Can have their progress determined and tracked, their expected costs reasonably established, their resource needs estimated
 - Are assignable to **one person**



Tools and Techniques

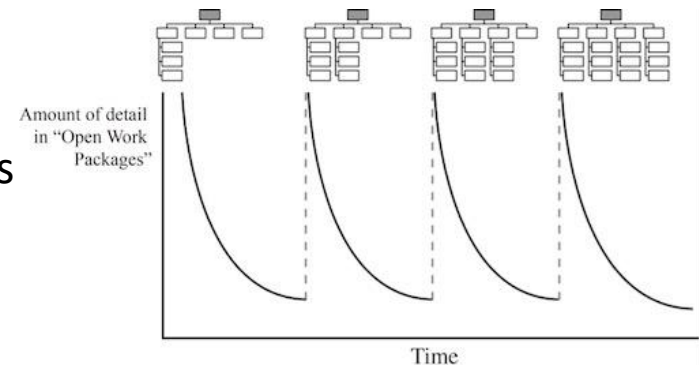
Rolling Wave planning

- An iterative planning technique in which the work to be accomplished in the near term is planned in detail, and the work further out is planned at a higher level.
 - Allows flexibility in dynamic or uncertain projects
 - Focuses effort where it's needed most (on upcoming work)
 - Adapts as more information becomes available



RWP & Milestone list

- Milestones Anchor the Rolling Plan
 - In RWP, detailed plans are created only for near-term work, while future work is outlined at a higher level.
 - Milestones serve as anchors or checkpoints in the high-level schedule.
- Milestones Trigger Planning Waves
 - As the project progresses and reaches a milestone, a new “wave” of planning is triggered.
 - Each wave increases the detail level for the next segment of work up to the next milestone.
- RWP Helps Define Milestones Realistically
 - Since RWP adapts as new information arises, it ensures that milestones are based on realistic scope and timelines, not just assumptions made at project initiation.



*Milestones give **clarity, accountability, and control**, while Rolling Wave Planning provides **flexibility and adaptability**. Used together, they allow you to **manage uncertainty without losing direction**.*

Tools and Techniques

Expert judgment

Ask somebody who has done this before to give an opinion on what activities will be needed to get the job done.



Meetings

You'll need to talk to the team to figure out what they think they need to do to achieve your project's goals. Getting the team together to discuss the plan will be a useful tool when you're defining the activities in your schedule.



Inputs

Project Management Plan

- **Schedule Management Plan**
 - Level of detail necessary to manage the work
- **Scope Baseline**
 - Project Scope Statement
 - WBS
 - WBS Dictionary

Enterprise environmental factors

- Project management information system (PMIS)

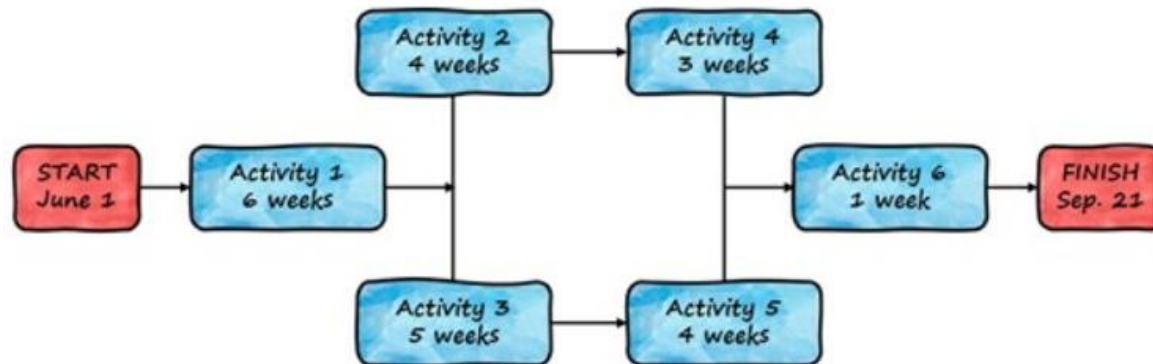
Organizational process Assets

- The scheduling methodology
- Lessons-learned knowledge base
- Previous similar projects

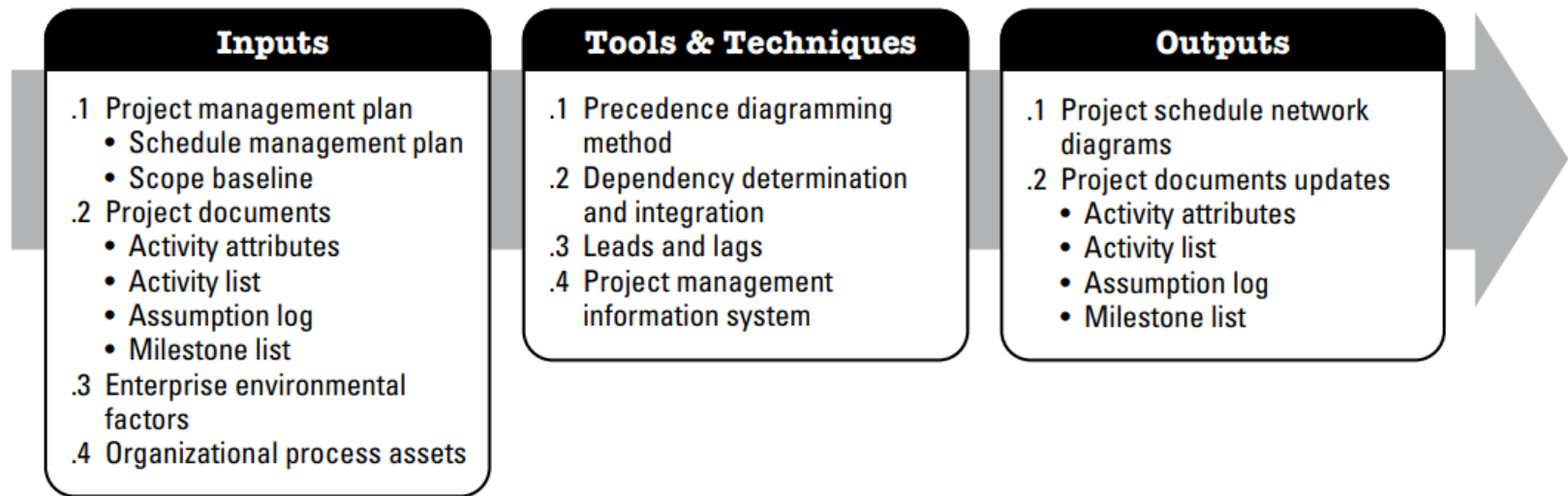
Sequence Activities

Sequence Activities

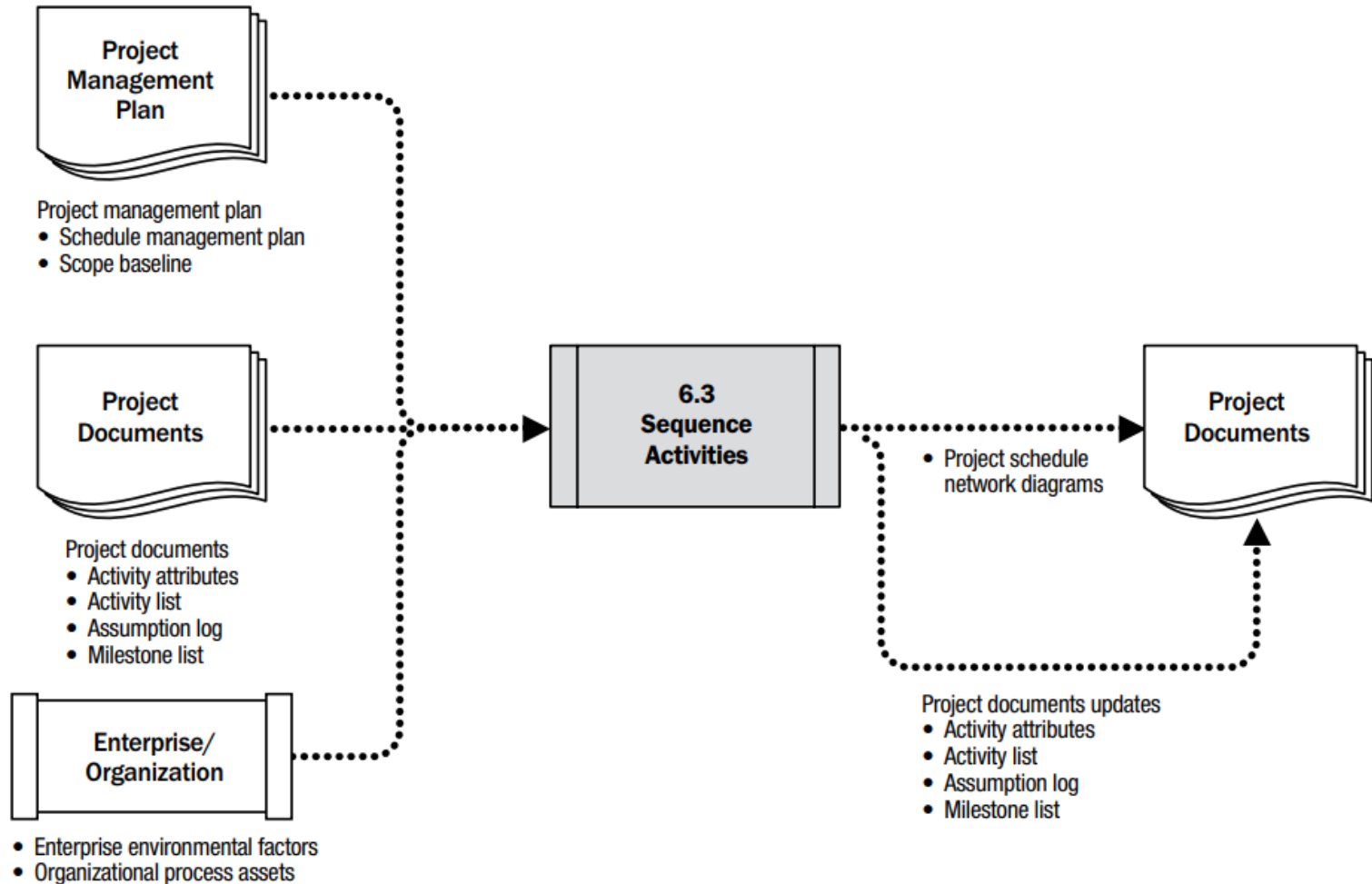
- **What:** process of identifying and documenting *relationships among the project activities*.
- **Why:** defines the logical sequence of work to *obtain the greatest efficiency* given all project constraints.
- **When:** performed throughout the project.
- **How:**
 - Determine dependencies between activities
 - Sequence activities by using appropriate diagramming technique
 - Applying leads and lags if necessary to support a realistic and achievable schedule



Process



Data Flow



Tools and Techniques

Dependency Determination

*A dependency is when **one activity relies on another** to start or finish.*

Category	Description	Example
Mandatory (Hard Logic)	Legally or technically required	"Foundation must be poured before building walls"
Discretionary (Soft Logic)	Based on best practices or team preference	"Write user manual after UI is finalized"
External	Outside the project team's control	"Wait for client feedback before proceeding"
Internal	Within project team's control	"Backend API must be ready before frontend integration"

Tools and Techniques

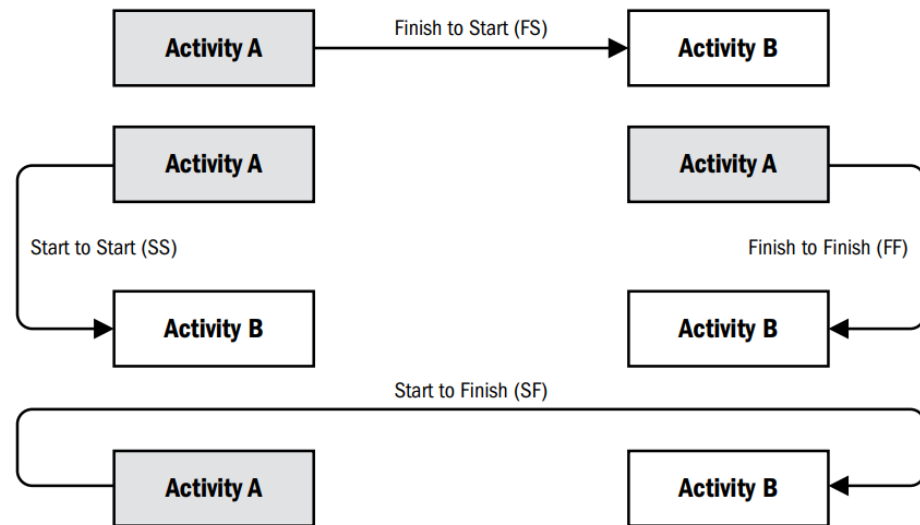
Precedence Diagramming Method (PDM)

4 logical relationships

- **Finish-to-start (FS):** most common type of relationship
 - **Activity A:** Predecessor
 - **Activity B:** Successor
- Finish-to-finish (FF)
- Start-to-start (SS)
- Start-to-finish (SF): very rarely used

Not recommend:

- Closed loop or Multiple relationships between the same activities



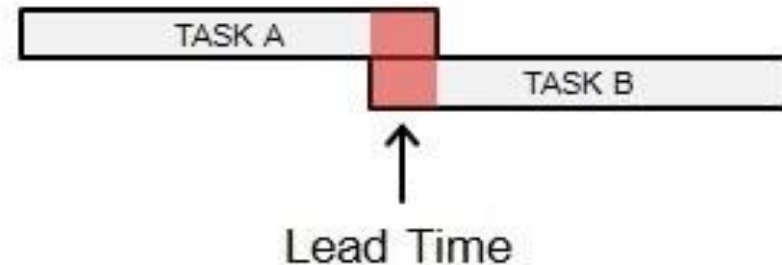
Type	Description	Example
Finish-to-Start (FS)	Task B cannot start until Task A finishes	"Code must be written before testing starts"
Start-to-Start (SS)	Task B cannot start until Task A starts	"Design and writing documentation start together"
Finish-to-Finish (FF)	Task B cannot finish until Task A finishes	"Training ends when course development ends"
Start-to-Finish (SF)	Task B cannot finish until Task A starts (rare)	"Old system decommission finishes only when new system starts"

Tools and Techniques

Leads and Lags

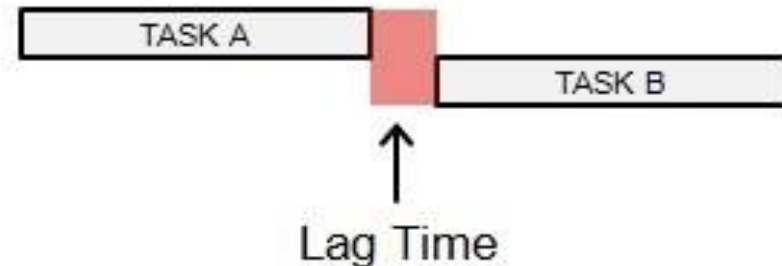
Lead

- The amount of time that a successor activity may be started prior to completion or predecessor.
- Ex: the landscaping could be scheduled to start 2 weeks prior to the scheduled punch list completion.



Lag

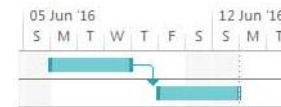
- The waiting time that a successor activity may have to wait after the completion of predecessor.
- Ex: construction only starts after finish foundation 3 days



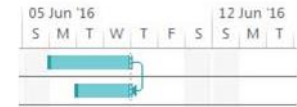
Tools and Techniques

Project management information system (PMIS)

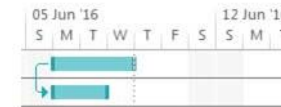
- Sequencing can be performed by using project management software or by using manual or automated techniques.
- Project management information systems includes scheduling software that has the capability to help plan, organize, and adjust the sequence of the activities; insert the logical relationships, lead and lag values; and differentiate the different types of dependencies.



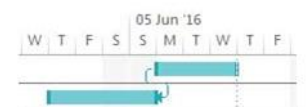
Finish-to-Start (FS)



Finish-to-Finish (FF)



Start-to-Start (SS)

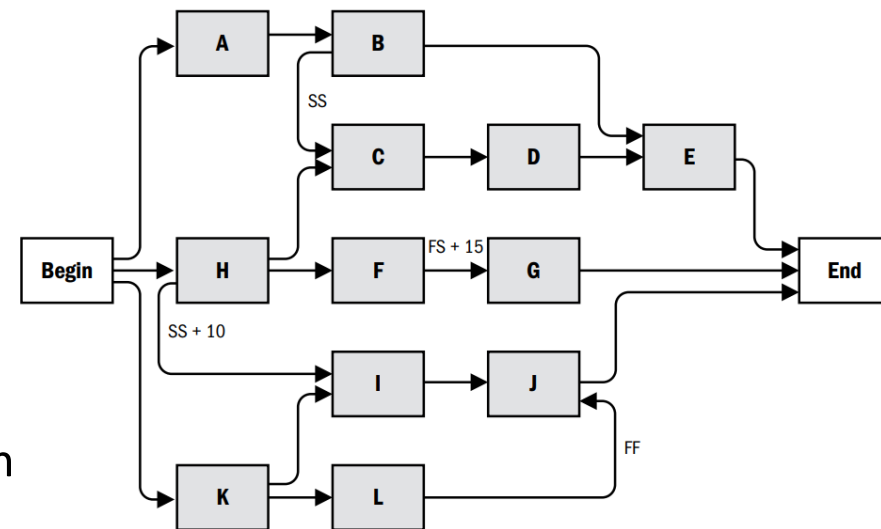


Start-to-Finish (SF)

Outputs

Project Schedule Network Diagrams

- Graphical representation of the logical relationships among the project schedule activities.
- Can include full project details, or have one or more summary activities.
- Activities with ***divergence*** and ***convergence*** are at ***greater risk*** as they are affected by multiple activities or can affect multiple activities.
 - Path convergence: multiple predecessor activities
 - Path divergence: multiple successor activities

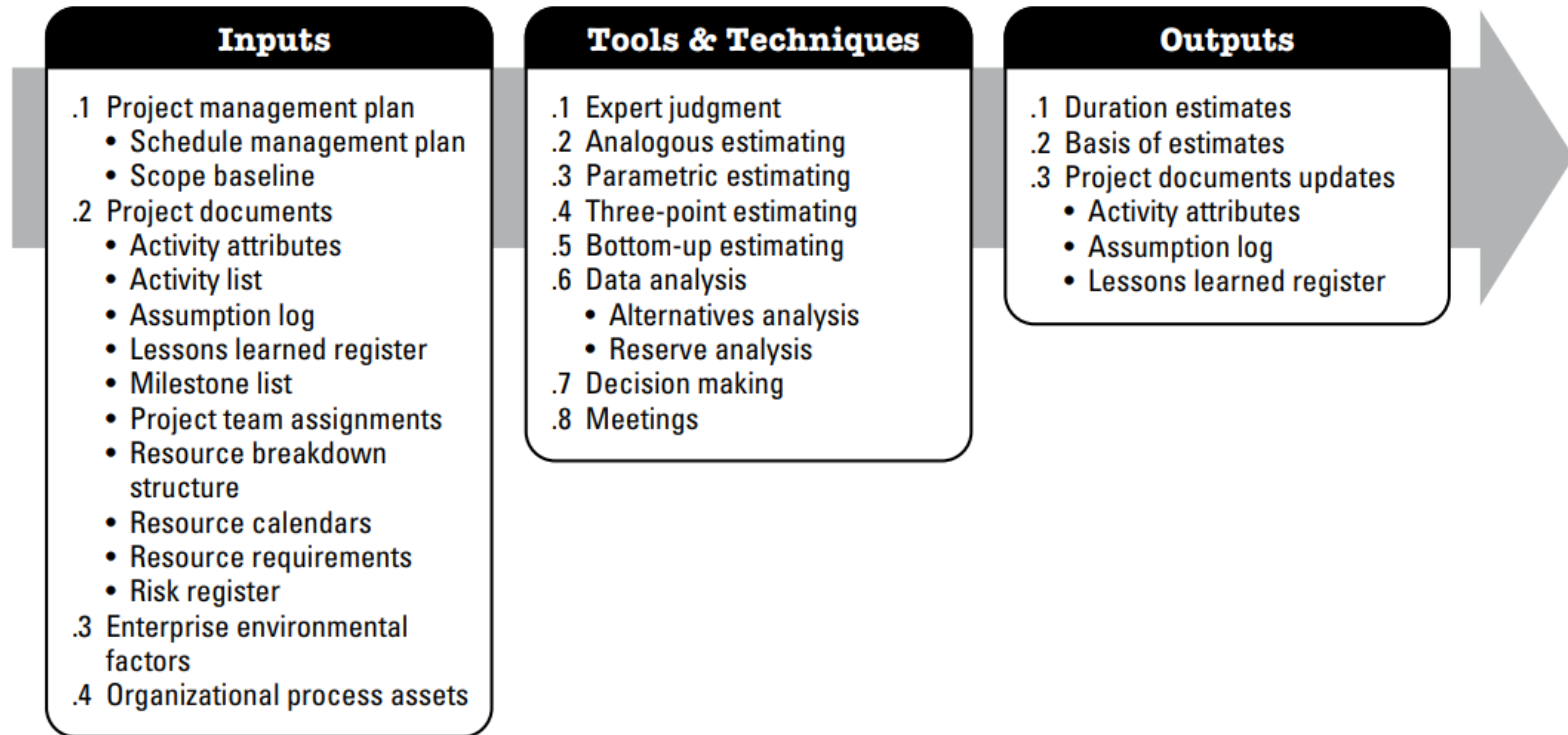


Estimate Activity Durations

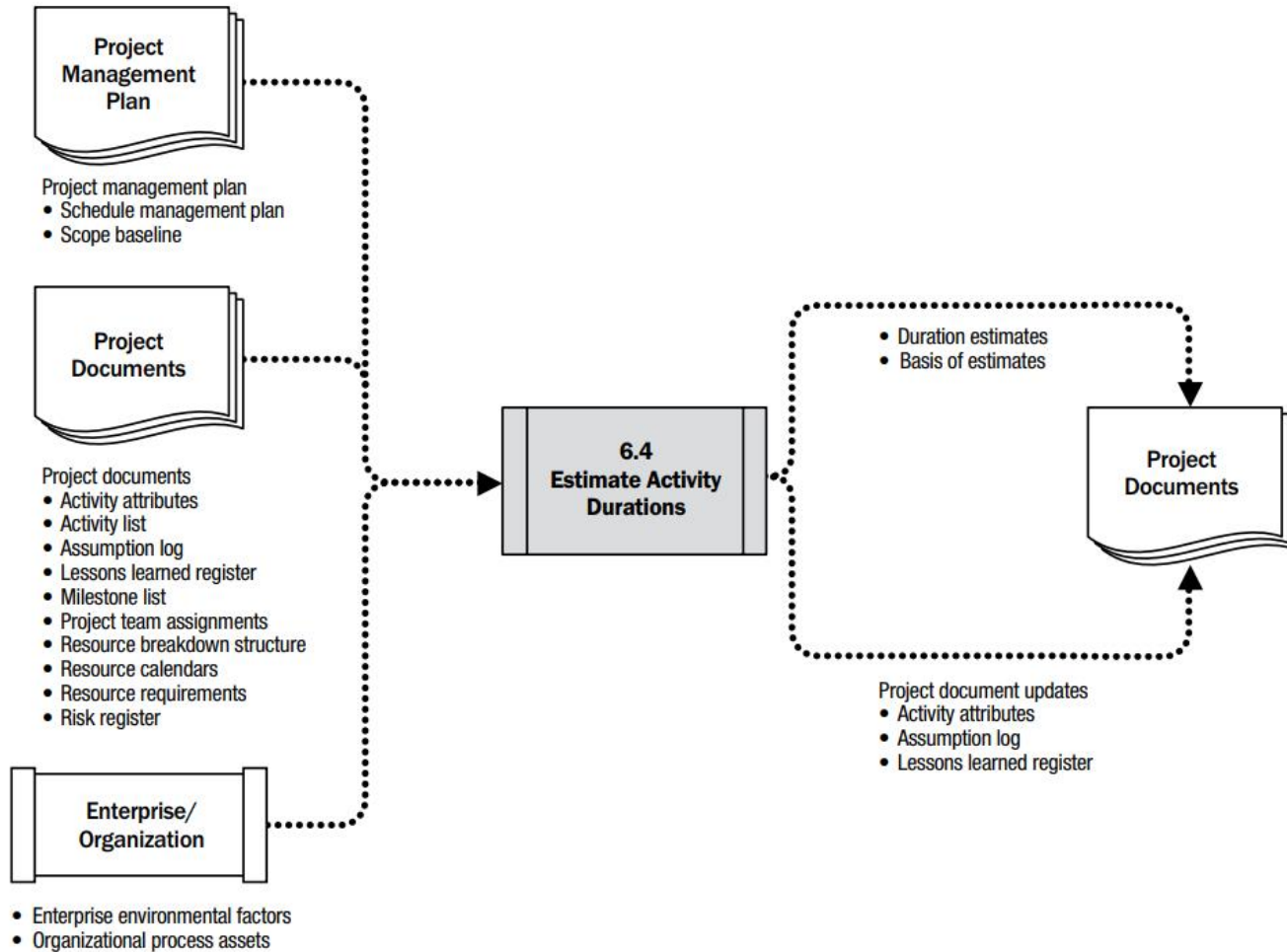
Estimate Activity Durations

- **What:** process of estimating the number *of work periods needed* to complete individual activities *with estimated resources*.
 - Project resources: can be material, people, equipment, or anything else capable of definition (usually other than labour) required for the completion of a project activity.
- **Why:** provides the amount of time each activity will take to complete.
- **When:** performed throughout the project.
- **How:**
 - Consider the given assumptions and constraints (quality and quantity of resources, motivation of staff, deadline, ...)
 - Estimate activity duration with appropriate techniques (base on number of resources, the skill proficiency of them,...)
 - Determine the uncertainty and estimate contingent reserve.
 - Re-evaluate and determine the optimal way to complete the activity if needed
 - Finalize and document the duration estimates

Process



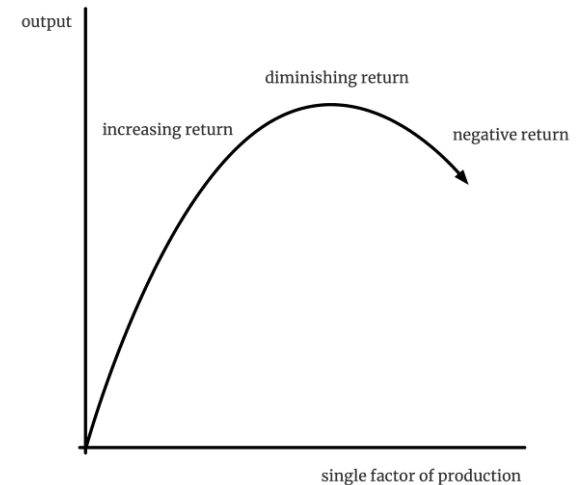
Data Flow



Affect factors to duration estimation

LAW OF DIMINISHING RETURNS

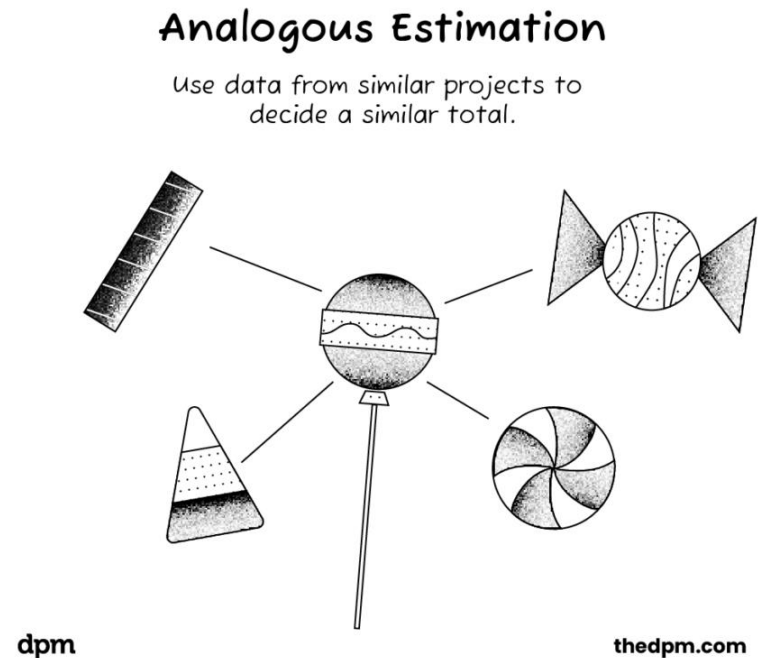
- **Law of diminishing returns:** After a certain point, adding more resources to a task will result in smaller and smaller gains in productivity—and may even make things worse
- **Advances in technology:** advances in technology may impact duration and resource needs
- **Motivation of staff:**
 - **Student Syndrome:** Waiting until the latest possible time to start (procrastination)
 - **Parkinson's law:** Work expands to fill available time



Tools & Techniques

Analogous Estimating

- Uses information from a previous, similar project, such as duration, budget, size and complexity for future project.
- Analogous estimate is generally **less costly and time consuming but generally less accurate**.
- This estimate will be more accurate if previous project is similar in nature and not just in appearance.
- Analogous duration estimating is frequently used to estimate project duration when **there is a limited amount of detailed information** about the project.



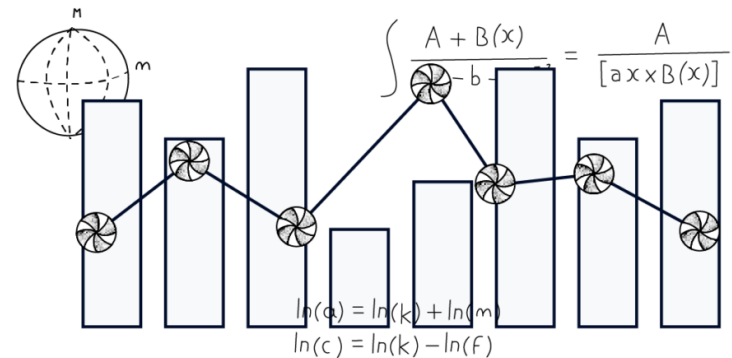
Tools & Techniques

Parametric Estimating

- Parametric estimate uses a **statistical relationship** between historical data and other variables.
 - Identify a measurable parameter (e.g., cost per unit, hours per line of code, dollars per square meter).
 - Use a formula or rate based on historical data or benchmarks.
 - Multiply the parameter by the quantity needed for the current project.
- Example - A resource will take 20hrs per module and hence 50 modules will take 1000hrs (50X20 = 1000hrs)

Parametric Estimation

Using data and variables to calculate the total.



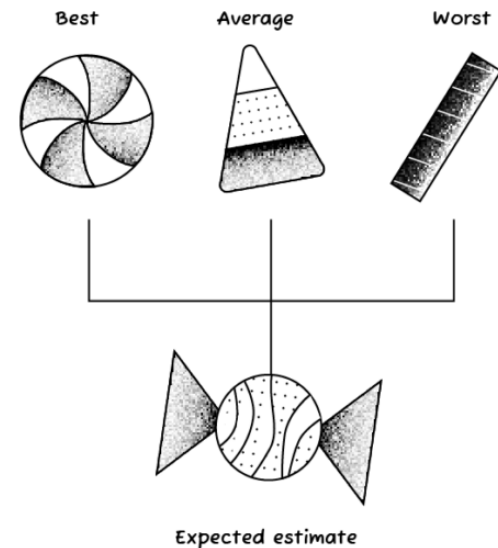
Tools & Techniques

Three point estimates (PERT)

- When there is insufficient historical data or when using judgmental data.
- A three-point estimate uses average of optimistic, most likely, and pessimistic estimates and hence improving the accuracy.
 - **Pessimistic (tP)** assumes the worst case scenario
 - **Most likely (tM)** – The realistic and most likely estimate
 - **Optimistic (tO)** is the best case scenario.
- $tE = (tO + tM + tP)/3$

Three Point Estimation

Get three different estimates and take an average to calculate the total.



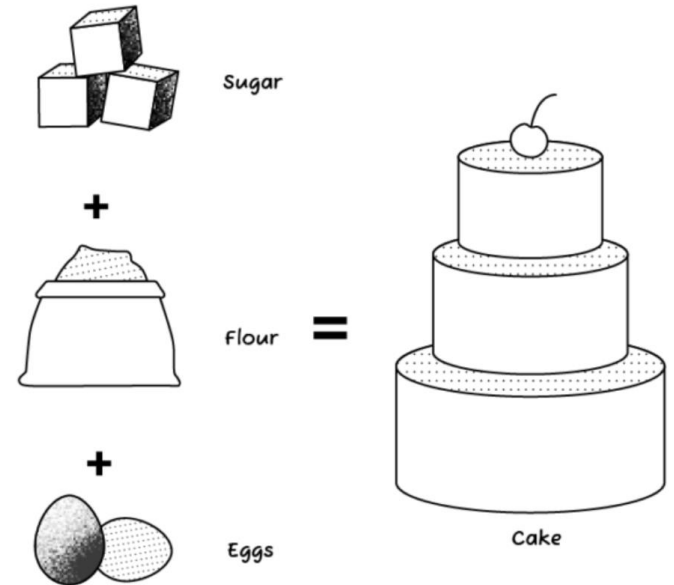
Beta distribution

$$\left(\text{Optimistic duration} + 4 \text{ Most likely duration} + \text{Pessimistic duration} \right) \div 6 = \text{Expected duration}$$

Tools & Techniques

Bottom-up Estimating

- The work within the activity is decomposed into more detail.
- Once these estimations have been performed, the pieces may be summed up from the bottom back to activity level.
- The most accurate



Tools & Techniques: Data analysis

Reserve Analysis

- Determine the amount of reserve needed for the project.
 - Contingency: allocated for identified risks that are accepted (known-unknowns)
 - Management: reserved for unforeseen work that is within scope of the project (unknown-unknowns)
- Contingency Reserve time —also called **time reserves**
 - percentage of time
 - number of work periods
- Contingency reserve shall be clearly identified and documented for future analysis

Alternatives analysis.

- Many activities can be completed in different ways and using various resource allocations.
- Alternative analysis is used to choose the best way to complete an activity, which compares:
 - various levels of resource capability or skills;
 - different tools (manual versus automated);
 - make, rent, or buy decisions regarding the resources.

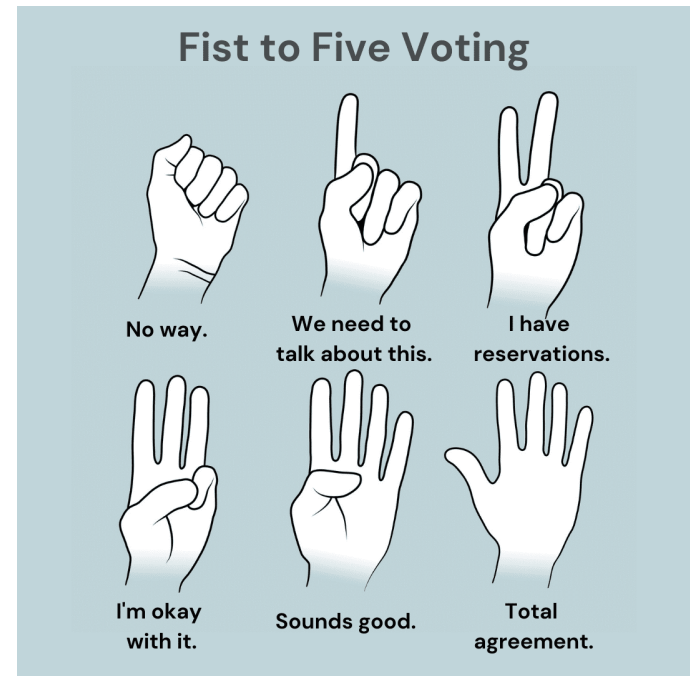
Tools & Techniques

Expert judgment will come from your project team members who are familiar with the work that has to be done. If you don't get their opinion, then there's a huge risk that your estimates will be wrong!

Decision-making techniques help the team decide on the best estimates for the activities they've defined.

Meetings help teams to work together when they estimate the work.

- In Agile method, sprint or iteration planning meetings to discuss prioritized product backlog items (user stories)



Management Reserve

- Keep It Outside the Schedule Baseline
- Build It into the Overall Project Delivery Date (Total Timeframe)
 - The project team commits to a schedule baseline (e.g., 8 months).
 - Senior management sets a project delivery deadline (e.g., 9 months).
 - The 1-month gap is your management reserve.
- Maintain a separate management reserve register.
- Using management reserve typically needs:
 - A formal change request
 - Approval from the project sponsor or steering committee

Element	Included in Schedule Baseline?	Controlled by	Purpose
Planned Work	Yes	Project Team	Execution & tracking
Contingency Reserve	Yes	Project Manager	Buffer for known risks
Management Reserve	No	Senior Management	Buffer for unknown risks

Outputs

Activity duration estimates

- Quantitative assessments of the likely number of time periods that are required to complete an activity, a phase, or a project.
- Do not include any lags
- May include some indication of the range of possible results. Ex:
 - A range of 10 days $\pm$ 2 days: activity will take at least 8 days and not more than 12
 - A 15% probability of exceeding 3 weeks, which indicates a high probability—85%—that the activity will take 3 weeks or less.

Basis of estimates

Additional details supporting the duration estimate (documentation)

- The basis of the estimate (i.e., how it was developed),
- All assumptions made,
- Any known constraints,
- Indication of the range of possible estimates (e.g., $\pm 10\%$) to indicate that the duration is estimated between a range of values)
- Indication of the confidence level of the final estimate

DURATION ESTIMATING WORKSHEET

Project name:

Project ID:

Project manager:

Project Sponsor:

PARAMETRIC ESTIMATES

Estimate ID	WBS ID	Task/Activity ID	Effort hours	Resource quantity	% Available	Performance factor	Duration estimate (hours)
<i>[unique ID for this estimate]</i>	<i>[WBS ID of activity]</i>	<i>[Task or activity ID]</i>	<i>[Effort hours]</i>	<i>[No. resource e.g. 2 people]</i>	<i>[% time available per day or week e.g. 75%]</i>	<i>[Average PF = 1.0. More skilled the lower the factor e.g. 0.8. Less skilled higher e.g. 1.2]</i>	<i>[Effort/(quantity x percent available x PF) = duration estimate]</i>
1-3P	1.2.1.3	26	150.00	2.00	75.0%	0.8	125

Parametric estimates

Analogous estimates

Three-point estimates

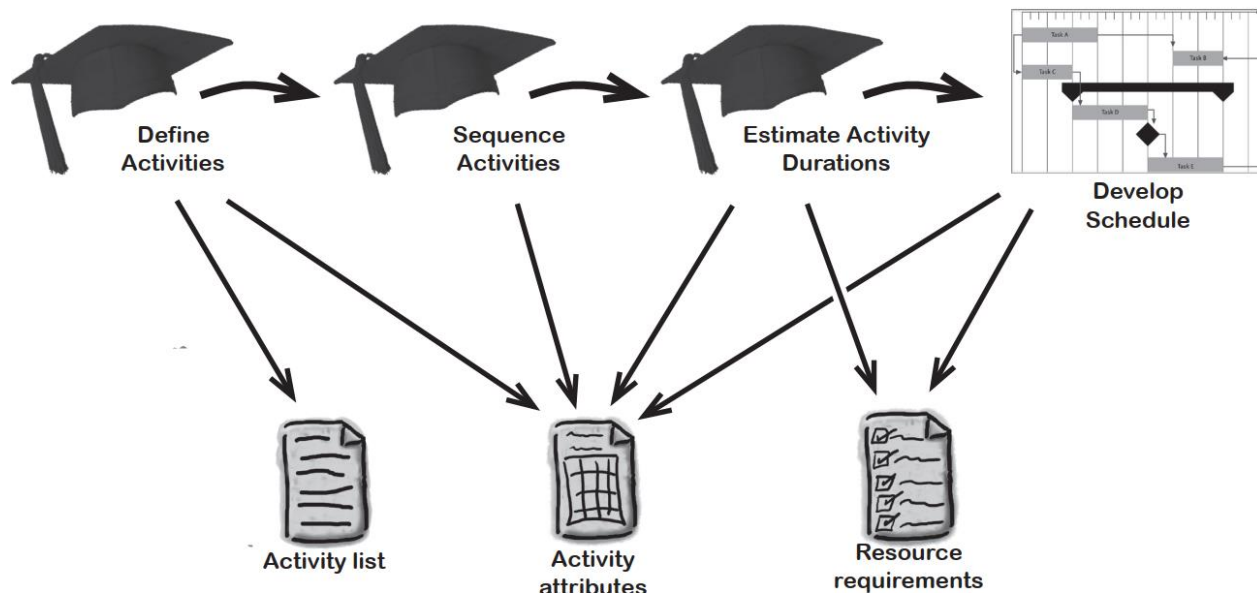
Useful res ...



Develop Schedule

Develop Schedule

- **What:** process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create a schedule model for project execution and monitoring and controlling. It generates a schedule model with planned dates for completing project activities.
- **Why:** serve as a baseline to track progress
- **When:** performed throughout the project (iterative process).

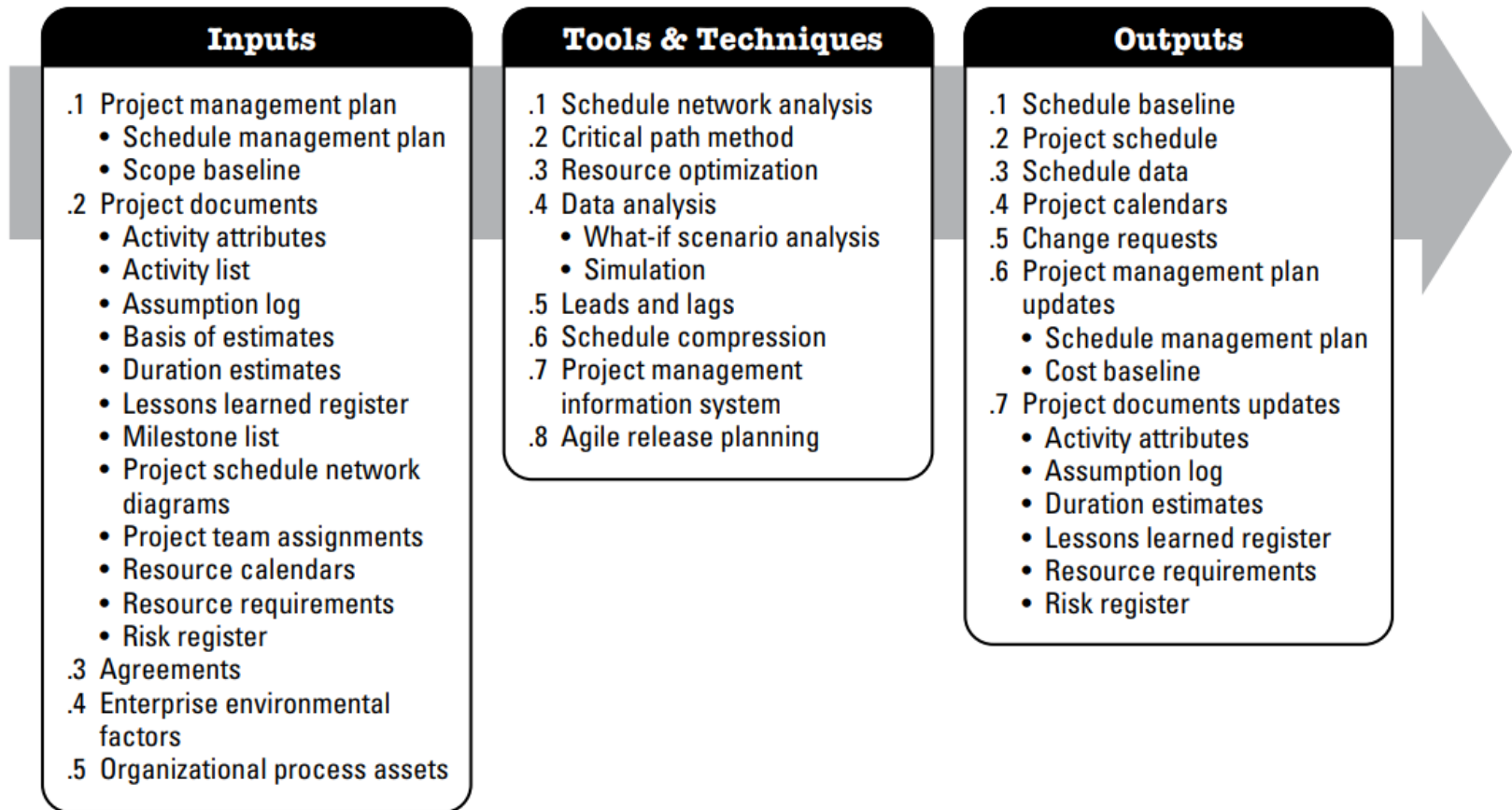


Develop Schedule

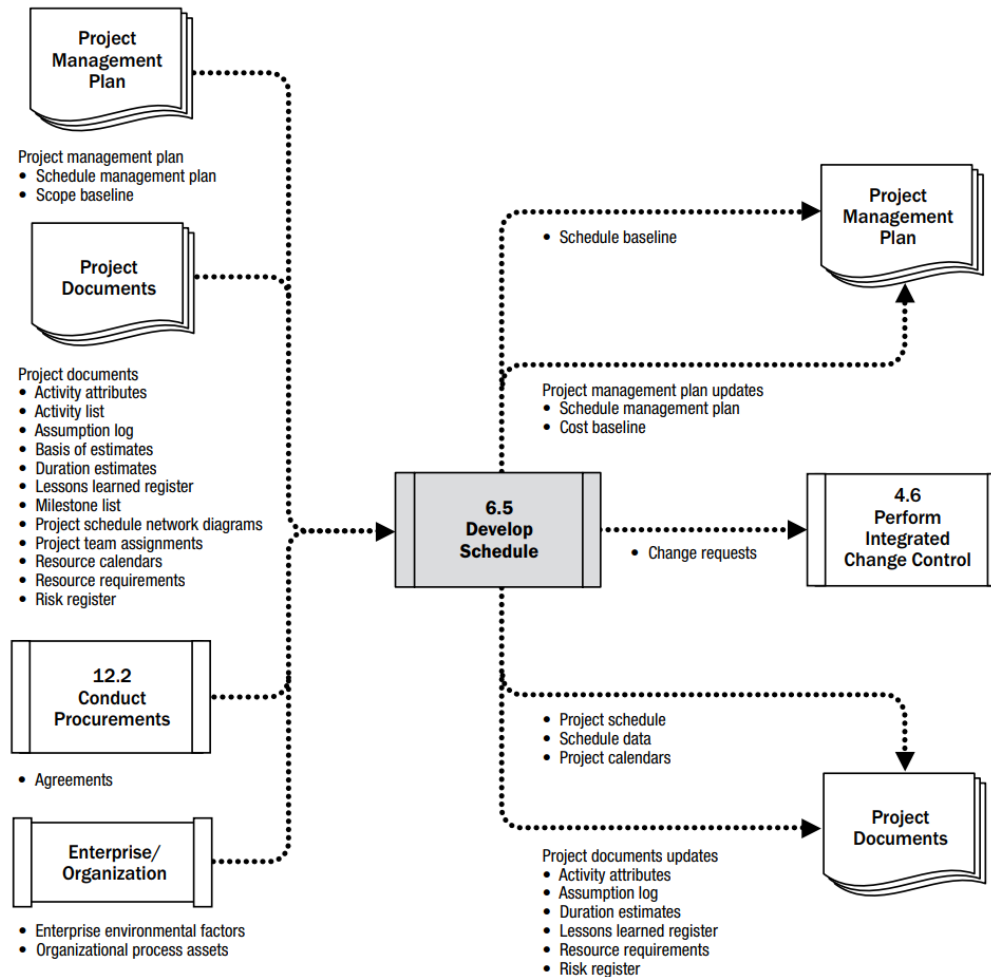
How?

- Once the project start or finish dates have been imposed , it is common to have the project staff assigned to the activities review their assigned activities. The project schedule is then analyzed to:
- Estimate the minimum project duration (**critical path**) and determine the amount of schedule flexibility.
- Determine conflicts with resource calendar and if **resource optimization** is required
- Determine conflict with milestones and if **schedule compression** is required
- Review and revise the duration estimates, resource estimates, and **schedule reserves** needed for risks
- Establish an approved project schedule that can serve as **a baseline** to track progress.

Process



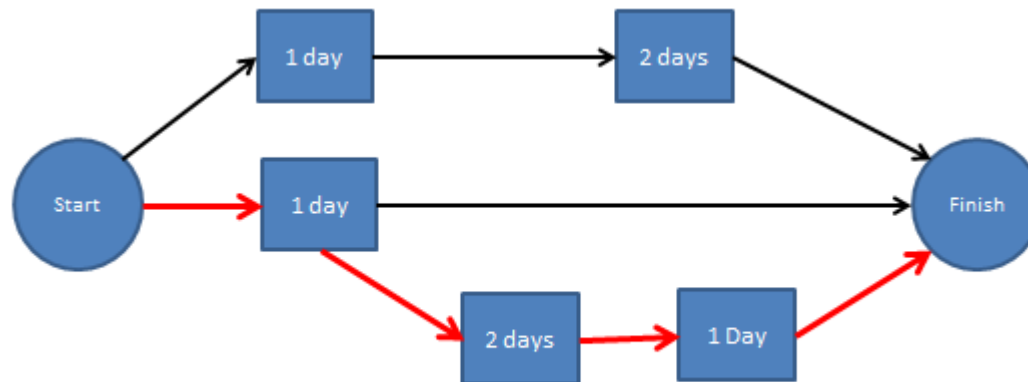
Data Flow



Tools & Techniques

Critical Path Method (CPM)

- CPM is used to calculate the **critical path(s)** and the amount of schedule flexibility (**total float**).
- Often activities happen in parallel, so the overall project duration is not the sum of all the activity estimates.
- The **critical path** is **the longest path** (by activity durations) through the project network diagram, also represents the **shortest time** the project can be finished in (any delays to activities on this path will delay the Finish date).



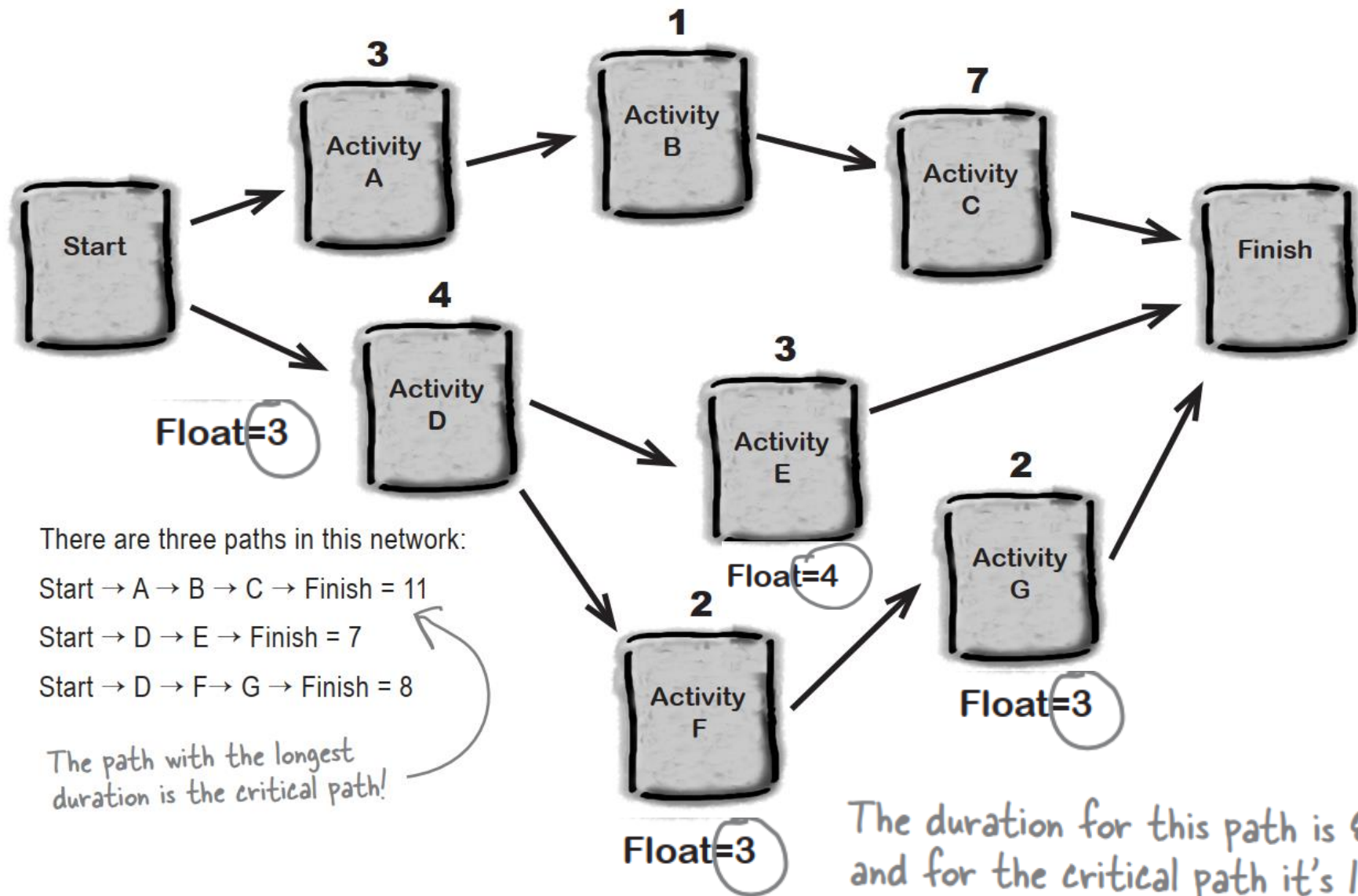
Critical Path = 4 Days

Critical Path Method

Schedule flexibility (float) determination

- **Free Float:** The amount of time an activity can be delayed **without delaying the early start** of any **immediate successor**.
- **Total Float:** The **maximum amount of time** an activity can be delayed **without delaying the project's finish date**.
- Activity on the critical path -> **float of zero**.

$$\begin{aligned}\text{Total Float} &= \text{Late Start (LS)} - \text{Early Start (ES)} \\ &= \text{Late Finish (LF)} - \text{Early Finish (EF)}\end{aligned}$$



Early/late start & early/late finish

Network Diagram Nodes

- ES = Earliest date an activity can be started
- EF = Earliest date an activity can be completed
- LS = Latest date an activity can start
- LF = Latest date an activity can be completed

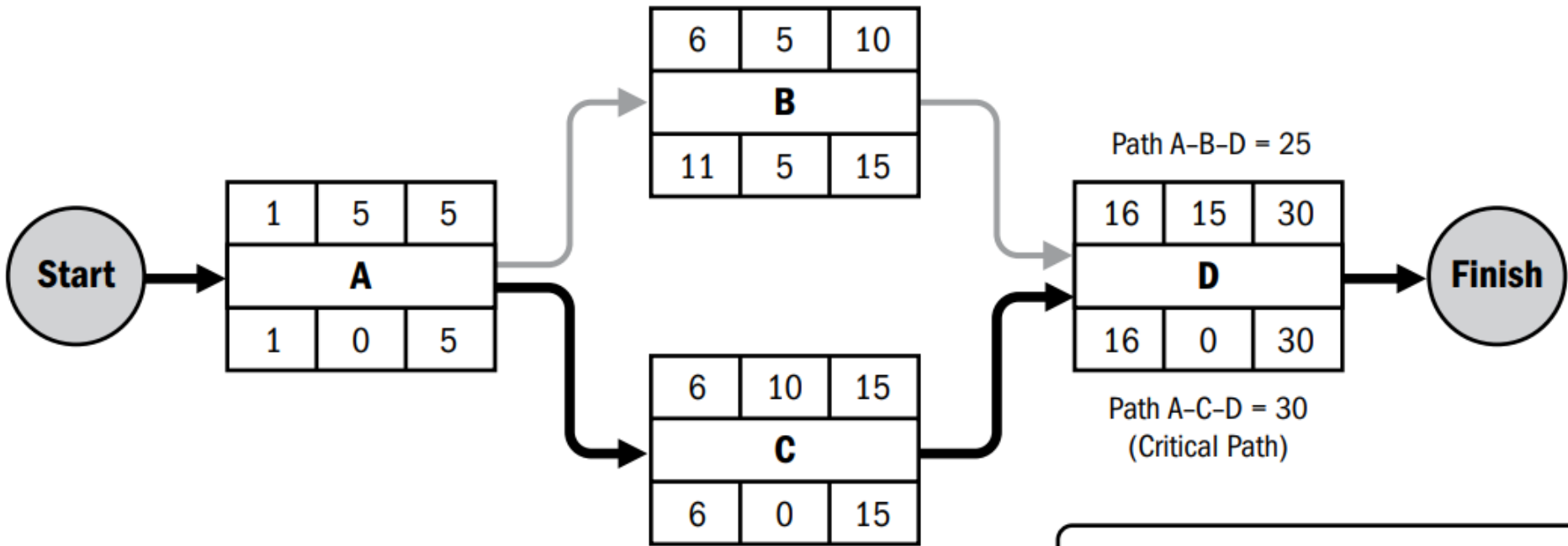
Early Start	Duration	Early Finish
Activity Name		
Late Start	Total Float	Late Finish

Early Start & Early Finish Calculation

- **Forward Pass:** to determine the earliest start and finish date
- **Early Finish (EF) = Early Start (ES) + Duration - 1**

Late Start & Late Finish Calculation

- **Backward Pass:** to determine the latest start and finish date
- **Late Start (LS) = Late Finish (LF) – Duration + 1**



NOTE: This example uses the accepted convention of the project starting on day 1 for calculating start and finish dates. There are other accepted conventions that may be used.

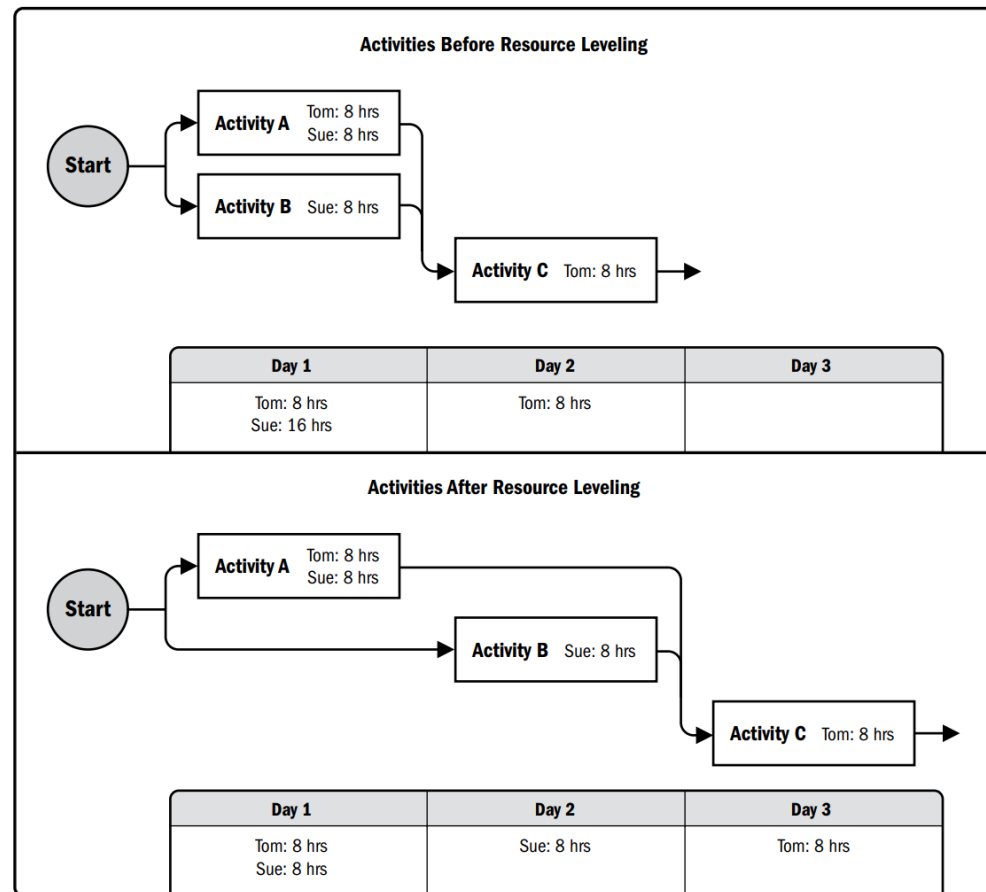
Critical float notice

- Total Float on this CP may be **positive** or **negative** (zero is the **normal** and expected condition)
 - **Positive:** The project **has built-in buffer time** or slack in the critical path (unusual and rare).
 - **Negative:** the project is already behind schedule based on **constraints** or **deadlines**.
 - The project is in schedule risk or noncompliant with deadlines.
 - Requires schedule compression (e.g., crashing or fast tracking) or revisiting constraints.
- A project can have **multiple critical paths**
 - Higher risk: If any critical path is delayed, the whole project is delayed.
 - Makes project control and recovery planning more complex.
- A **near-critical path** is a sequence of activities that is **almost as long** as the critical path — usually has **low total float** (e.g., 1–2 days).
 - Not currently critical, but can become critical with minor delays.
 - High-risk areas for float erosion.

Tools & Techniques: Resource Optimization

Resource leveling

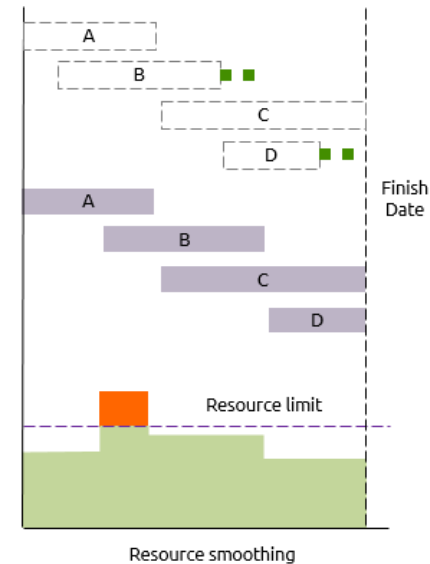
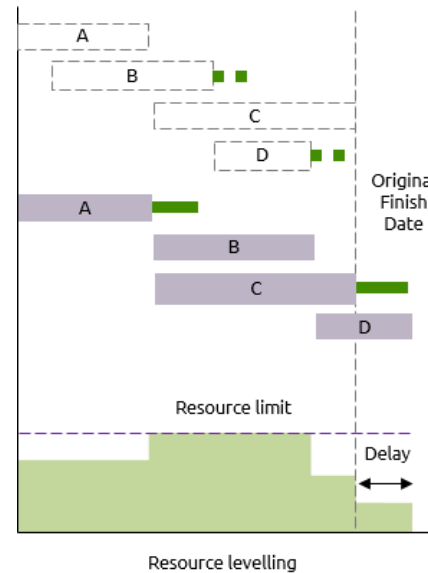
- Technique to adjust activities to **balance demand for resources**
- Used when **shared and critical resources** only available at certain times, or a resource assigned to two or more activities at the same time
- Project's critical path is changed, usually to increase



Tools & Techniques: Resource Optimization

Resource Smoothing

- Technique to adjust activities to meet **predefined resources limits**
- Activities **only be delayed within their free and total float**. Resource smoothing may not be able to optimize all resources
- Project's critical path is not changed and completion date may not be delayed



	Resource leveling	Resource smoothing
1	Resource limited scheduling technique; Importance is given to the limited resources	Time limited scheduling technique; Importance is given to the duration of the project
2	Removes all resource conflicts	Removes as much resource conflicts as possible; but, may not remove all resource conflicts
3	May not require additional resources	May require additional resources to address left over resource conflicts
4	Activities may be shifted beyond the float available while rescheduling the activities	Activities are shifted only to the extent of the float available
5	Generally, the project duration gets extended	The project duration remains the same
6	May change the critical path	No change in critical path

Tools & Techniques: Data Analysis

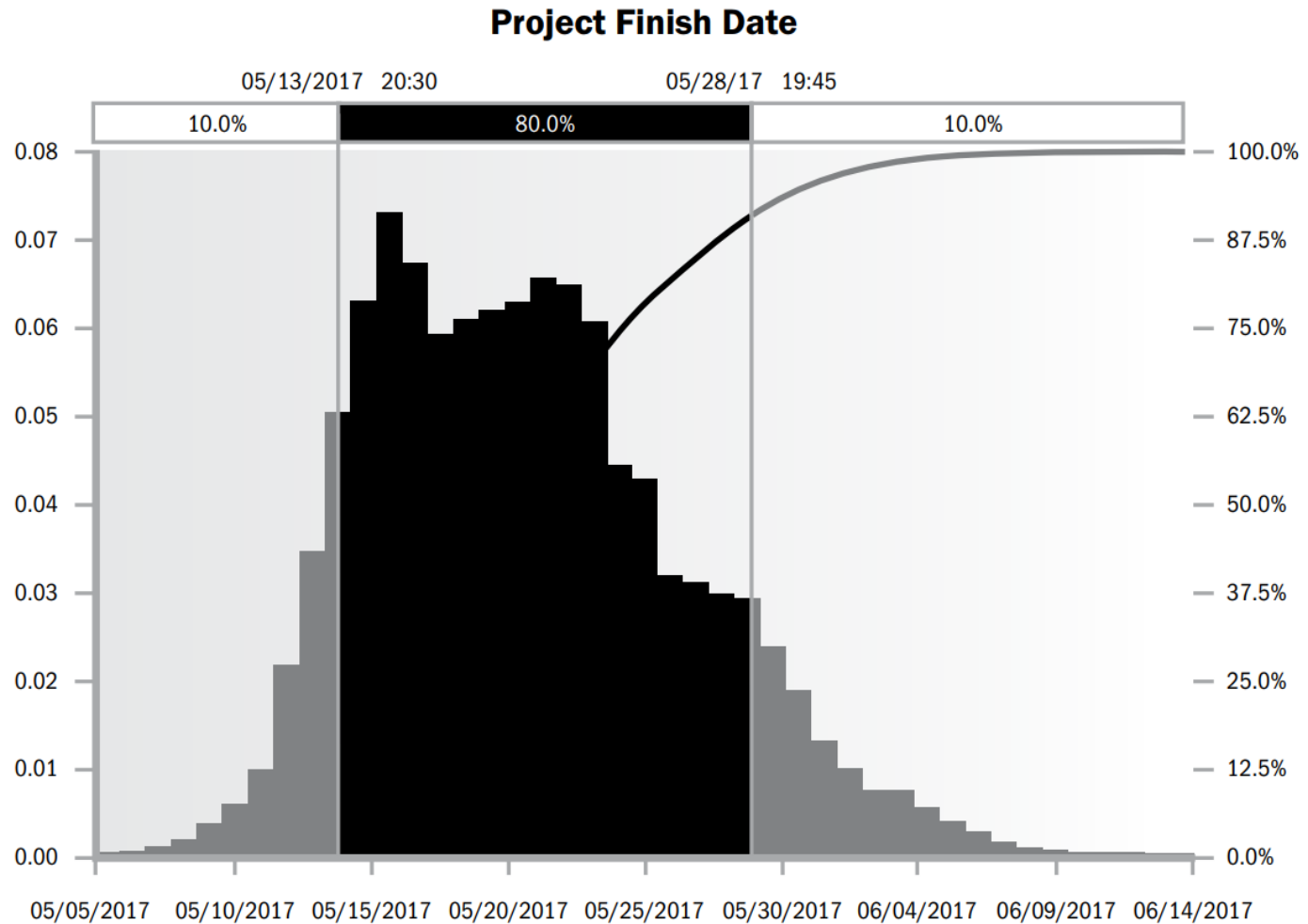
What-If Scenario Analysis

- A predictive technique used to evaluate the impact of specific hypothetical changes on the project schedule.
- What if the situation represented by scenario “X” happens?
- Used to access the **feasibility of the project schedule** under adverse conditions

Simulation

- Simulation models the combined effects of individual project risks and other sources of uncertainty to evaluate their potential impact on achieving project objectives.
- **Monte Carlo Simulation:** a quantitative risk analysis technique that uses probability distributions and simulations to analyze uncertainty in the schedule.

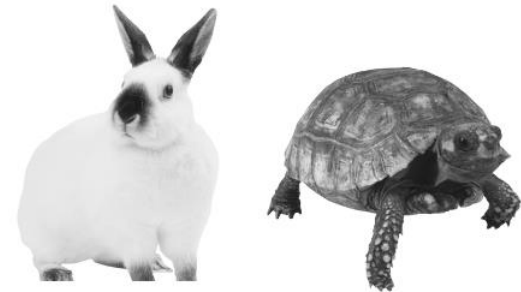
Tools & Techniques: Data Analysis



Tools & Techniques: Leads and Lags

Leads and Lags

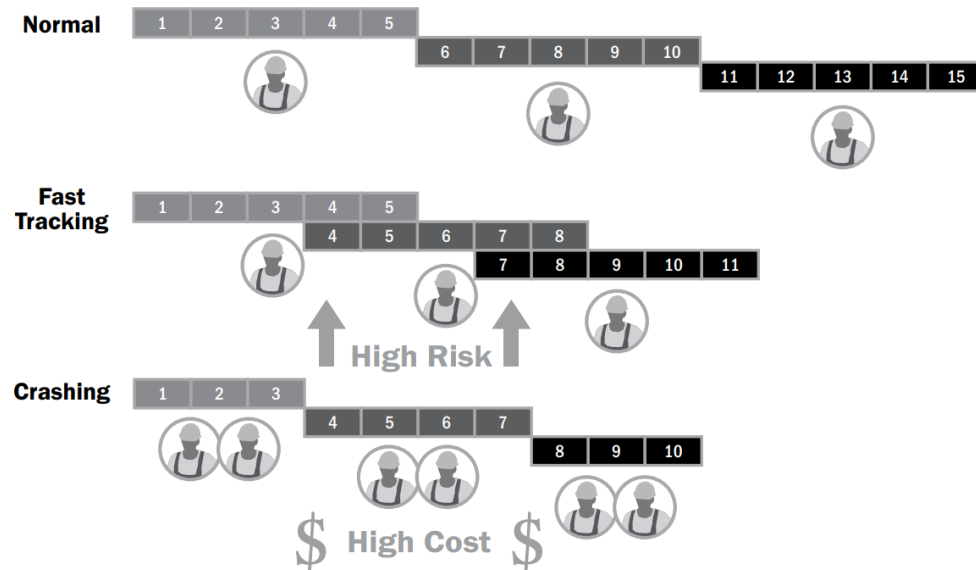
- Refinements applied during network analysis to develop a ***viable schedule*** by adjusting the start time of the successor activities
- Leads are used ***to advance a successor activity*** with respect to the predecessor activity
- Lags are used where processes require a set period of time to elapse between the predecessors and successors without work or resource impact.



Tools & Techniques: Schedule Compression

Shorten or accelerate the schedule duration without reducing the project scope in order to meet schedule constraints, imposed dates, or other schedule objectives.

- **Crashing:** adding resources (ex. approving overtime, bringing in additional resources, or paying to expedite delivery to activities on the critical path). => increased cost
- **Fast Tracking:** Critical activities that would normally be done in sequence are allowed to be **done in parallel** or with some overlap => may result in rework and **increases the risk**.
(Communication requirements increases during fast tracking).



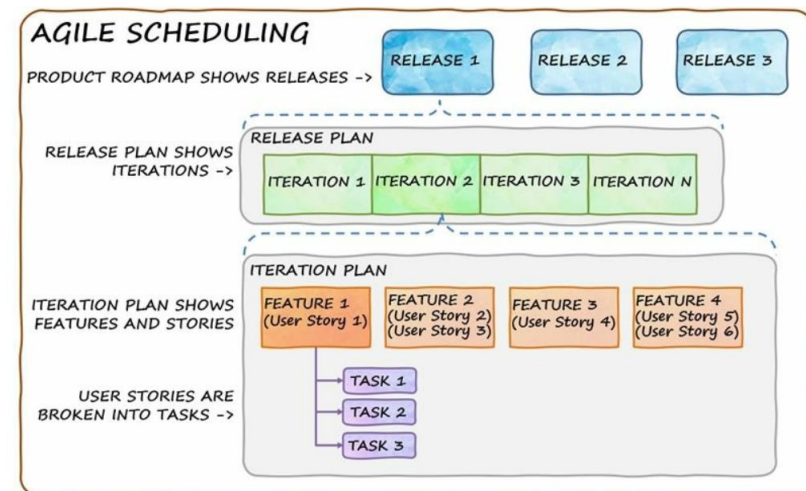
Tools & Techniques

Project management information systems

- Automated scheduling tool generates start and finish data of the activities based on the inputs of activities, network diagram, activity resources and durations.
- A scheduling tool can be used in conjunction with other project management software applications as well as manual methods

Agile release planning

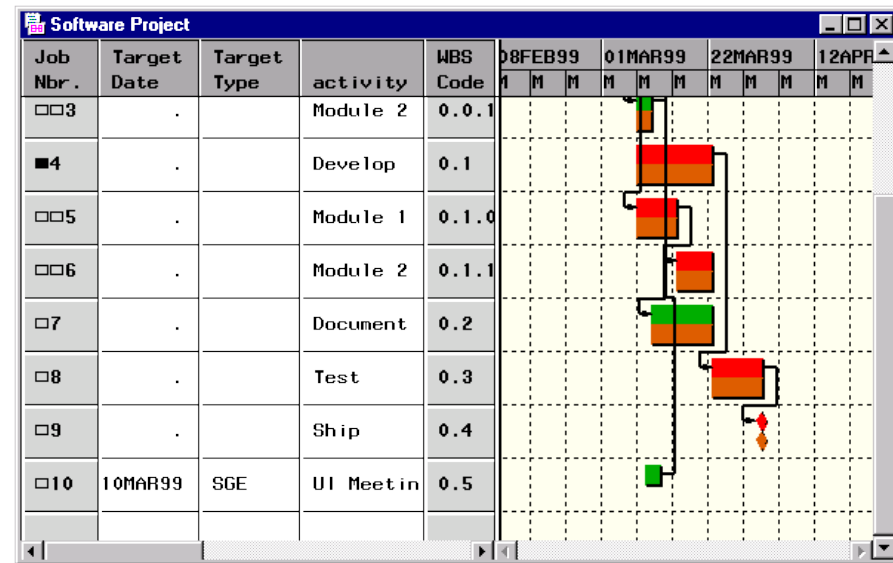
- An Agile release plan takes the list of user stories that have been estimated for a release and uses their estimated size of effort to map them to an upcoming iteration.
- The product owner reprioritizes stories and change the order in which stories will be released.



Outputs

Schedule Baseline

- A specific version of the project schedule developed from the schedule network analysis
- It is accepted and approved by the appropriate stakeholders, normally the project management team and functional managers.
- Can be changed only through formal change control procedures
- The schedule baseline is a component of the **project management plan**



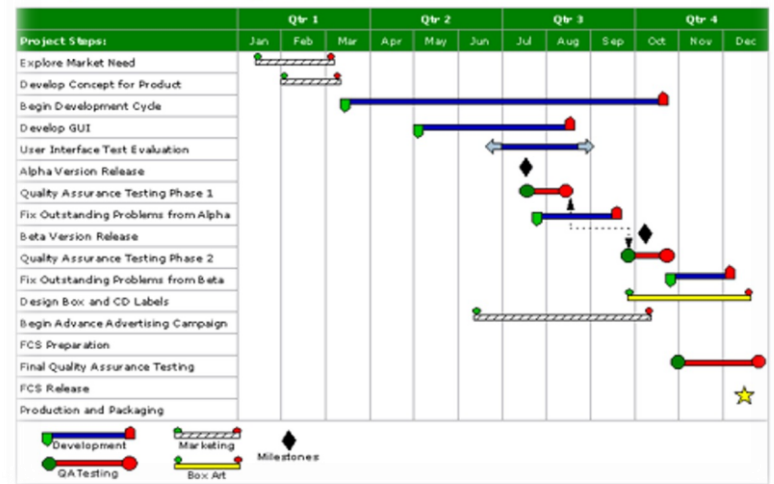
Outputs

Project Schedule

- Output of a **schedule model**, including planned dates, durations, milestones and resources
- At the beginning of project execution, the Project Schedule is the same as the Schedule baseline.
- As work is done on the project, the actual progress is updated on the Project Schedule. At any given date, the latest version of the actual (Project) Schedule is referred to as the “Project Schedule”.

- The schedule can be displayed in a variety of ways:
 - *Milestone Charts (or Master schedule)*
 - *Project Schedule Network Diagram (PDM)*
 - *Gantt Charts/ Bar Charts*

Project Development Schedule



Milestone Schedule

Activity Identifier	Activity Description	Calendar units	Project Schedule Time Frame				
			Period 1	Period 2	Period 3	Period 4	Period 5
1.1.MB	Begin New Product Z	0					
1.1.1.M1	Complete Component 1	0					
1.1.2.M1	Complete Component 2	0					
1.1.3.M1	Complete Integration of Components 1 & 2	0					
1.1.3.MF	Finish New Product Z	0					

← Data Date

Summary Schedule

Activity Identifier	Activity Description	Calendar units	Project Schedule Time Frame				
			Period 1	Period 2	Period 3	Period 4	Period 5
1.1	Develop and Deliver New Product Z	120					
1.1.1	Work Package 1: Component 1	67					
1.1.2	Work Package 2: Component 2	53					
1.1.3	Work Package 3: Integrated Components 1 and 2	53					

← Data Date

Detailed Schedule

Activity Identifier	Activity Description	Calendar units	Project Schedule Time Frame				
			Period 1	Period 2	Period 3	Period 4	Period 5
1.1.MB	Begin New Product Z	0					
1.1	Develop and Deliver Product Z	120					
1.1.1	Work Package 1: Component 1	67					
1.1.1.D	Design Component 1	20					
1.1.1.B	Build Component 1	33					
1.1.1.T	Test Component 1	14					
1.1.1.M1	Complete Component 1	0					

Schedule Baseline

High-Level, Approved Plan

- Approved **start and finish dates** for major tasks or milestones
- Overall **timeline and durations**
- **Milestone dates** and deliverables
- High-level **dependencies and constraints**
- Does **not** usually include:
 - Detailed resource assignments
 - Actual progress updates
 - Frequent revisions

Purpose:

- Basis for **performance measurement**
- Reference for **tracking variance**
- Documented in the **Project Management Plan**

Project Schedule

Detailed, Operational Tool

- All information from the schedule baseline **plus**:
 - **Detailed activities**
 - **Actual start and finish dates**
 - **Assigned resources**
 - **Calendars** (e.g., working days, shifts)
 - **Progress updates** (e.g., % complete)
 - **Revised estimates** and current forecasts

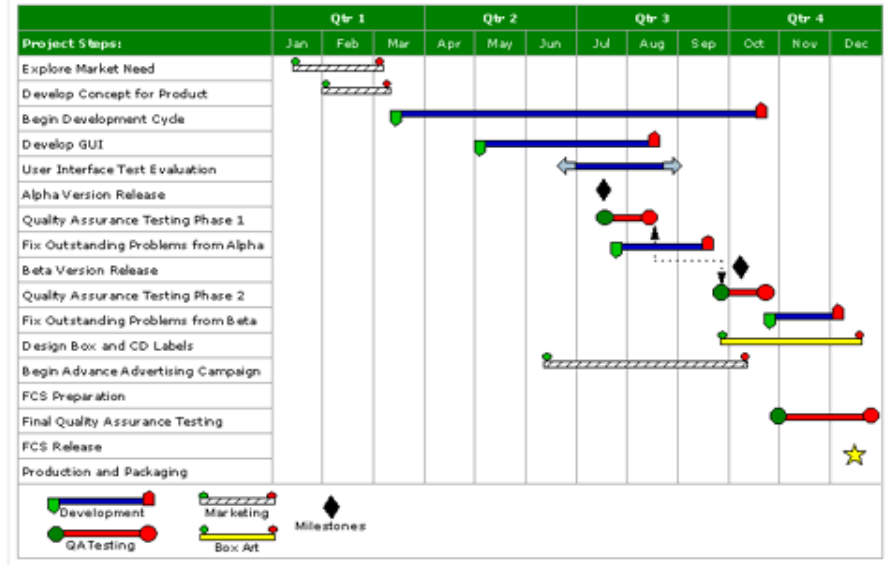
Purpose:

- Used by the team to **manage day-to-day work**
- **Frequently updated** as the project progresses

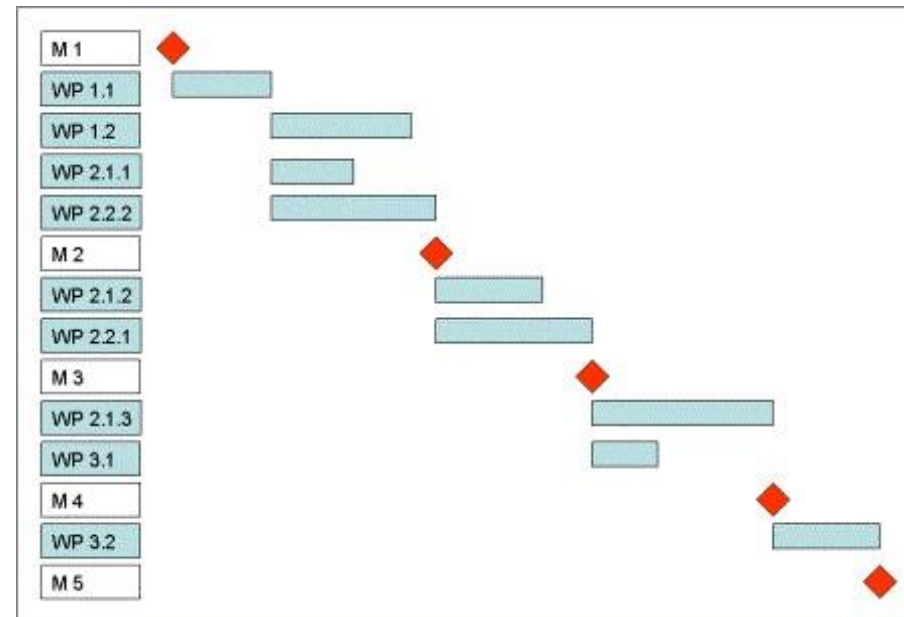
Element	Schedule Baseline	Project Schedule
Status of document	Approved (frozen)	Dynamic (living document)
Level of detail	High-level	Detailed
Includes actual dates	✗	✓
Includes resource assignments	✗ (usually not)	✓
Updated regularly	✗	✓
Used for reporting progress	✓ (for variance)	✓ (for day-to-day tracking)
Owned by	PM/Stakeholders	PM/Team

Gantt Charts/ Bar Charts

Project Development Schedule



Milestone Charts



Outputs

Schedule Data

- Milestones
- Schedule activities and activity attributes
- Identified assumptions and constraints
- Supporting Details: Resource Requirements, resource histograms, etc.

Project Calendar

- Working days and shifts available for schedule activities
- Like timetable, outlook calendar

Change request

- Modifications to the project scope or project schedule may result in change requests to the scope baseline, and/or other components of the project management plan.



Inputs

Project Management Plan

- Schedule Management Plan
- Scope baseline

Project documents

- Milestone list
- Activity attributes
- Activity list
- Assumption log
- Project team assignments
- Resource calendars
- Resource requirements
- Duration estimates
- Basis of estimates
- Project schedule network diagrams
- Lessons learned register
- Risk register

Agreement

- Detailed schedule developed by suppliers to meet contractual commitments

Enterprise environmental factors

- Government or industry standards, and
- Communication channels.

Organizational process assets

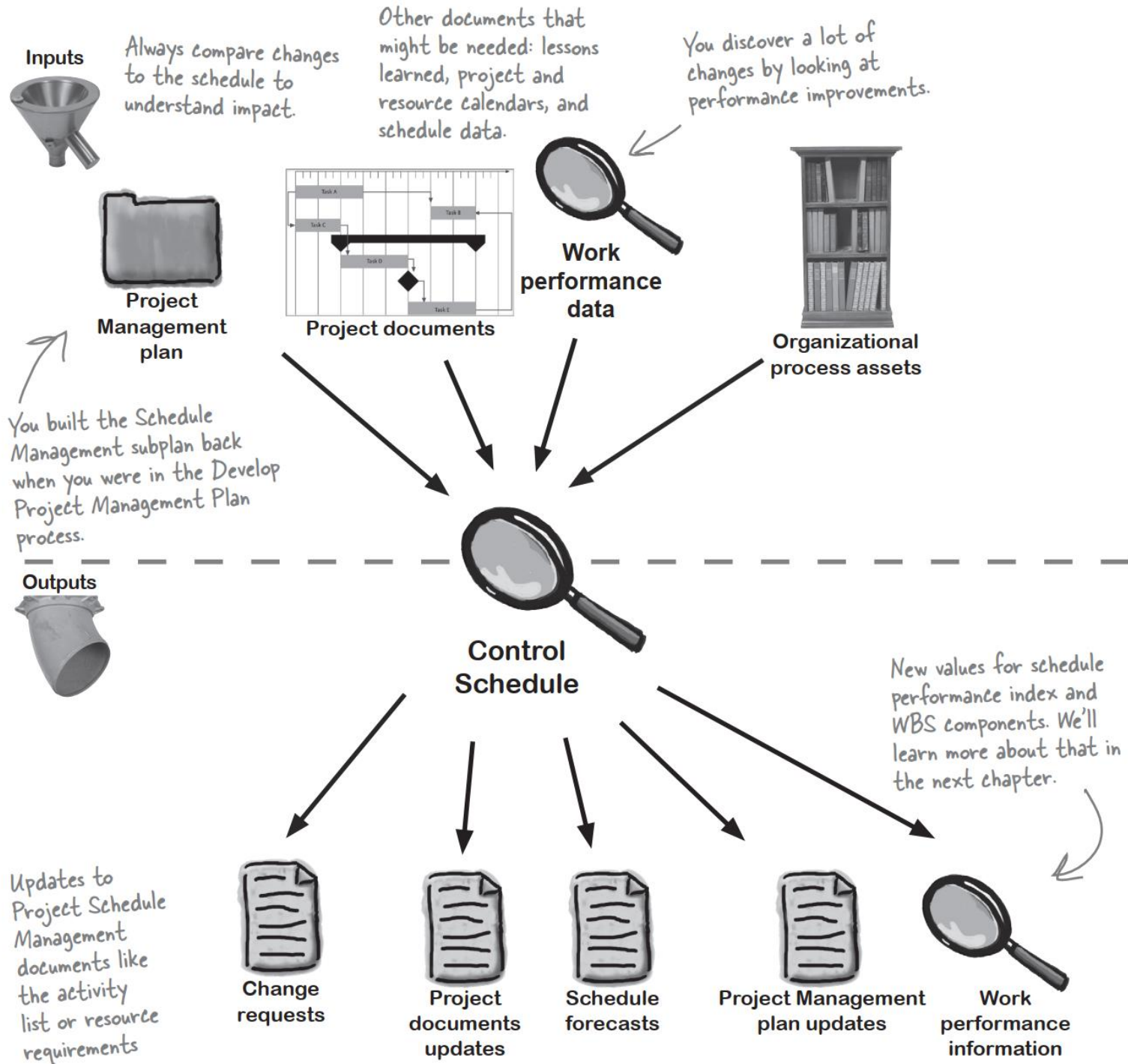
- Scheduling methodology containing the policies governing schedule model development and maintenance, and
- Project calendar(s).

Control Schedule

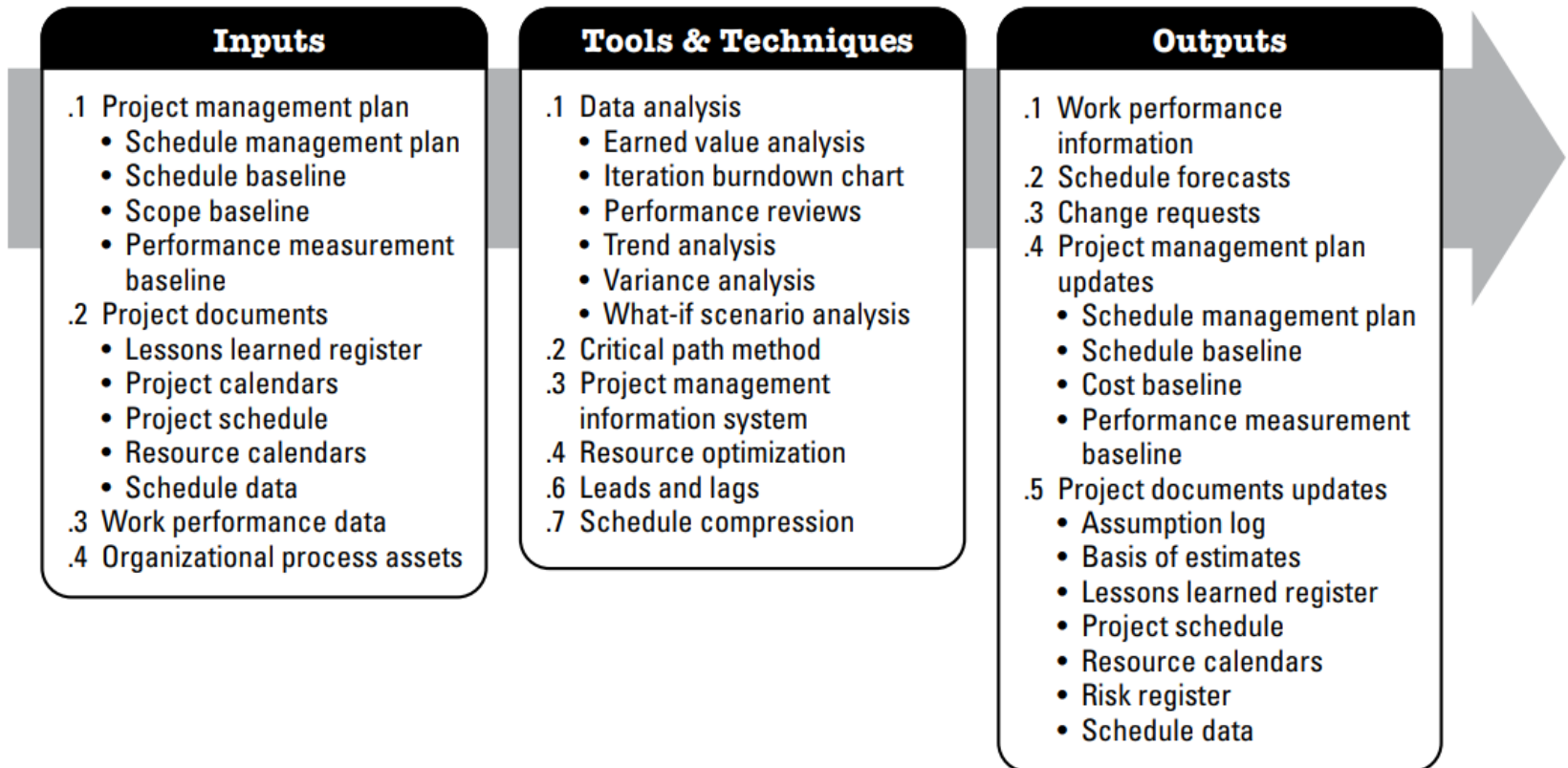
Control Schedule

- **What:** process of monitoring the status of the project to update the project schedule and managing changes to the schedule baseline.
- **Why:** assure schedule baseline is maintained throughout the project
- **When:** performed throughout the project (iterative process).
- **How:**
 - Determining the current status of the project schedule
 - Influencing the factors that create schedule changes
 - Reconsidering necessary schedule reserves
 - Determining if the project schedule has changed
 - Managing the actual changes as they occur.

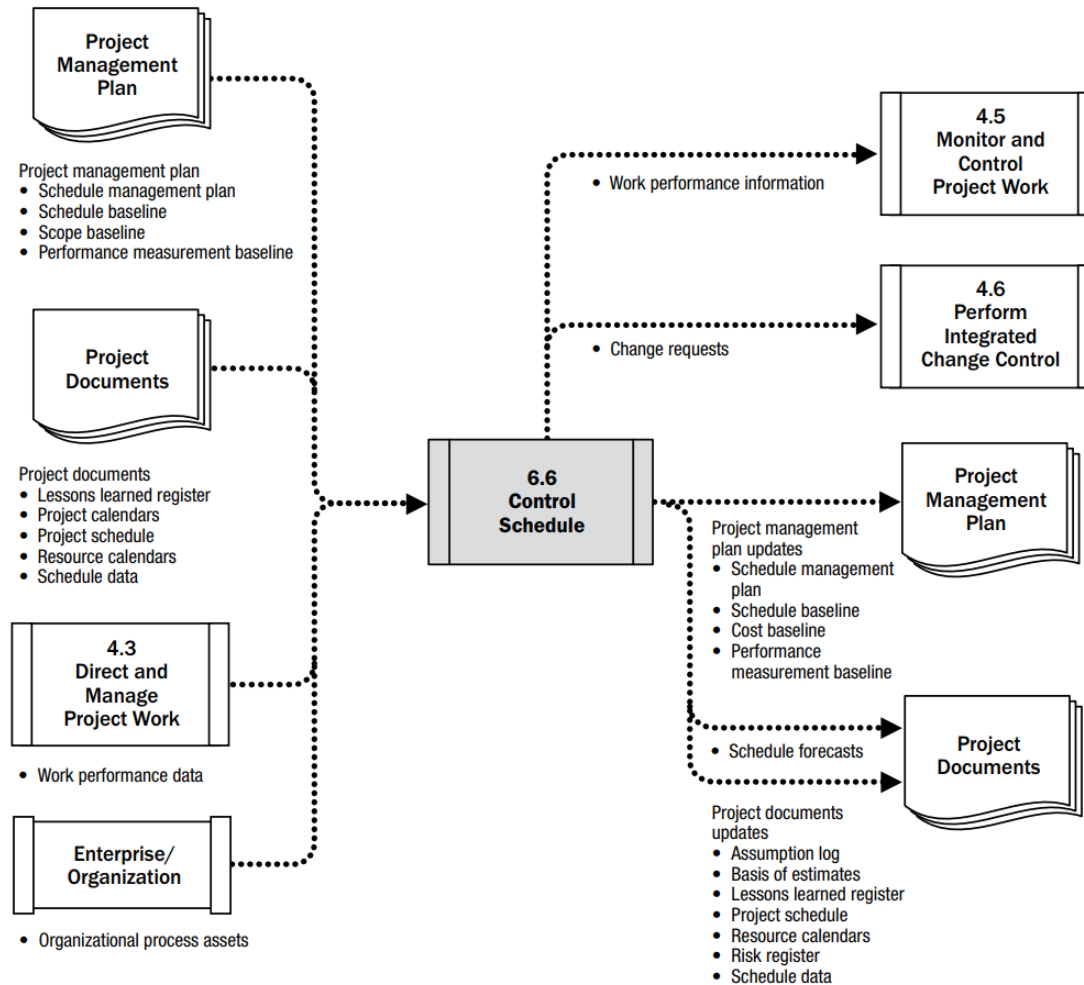




Process



Data Flow

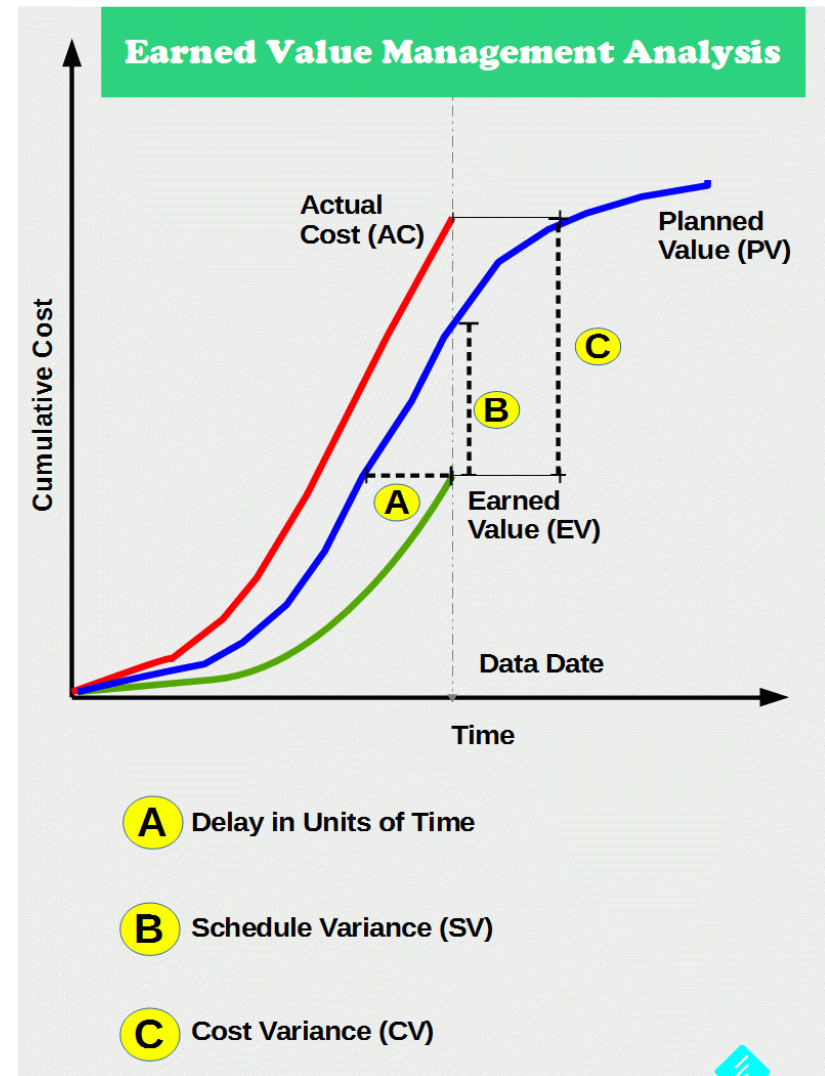


Tools & Techniques: Data analysis

Earned Value Analysis

Compares the performance measurement baseline to the actual schedule and cost performance.

- *Planned value (PV)*: authorized budget assigned to scheduled work.
- *Earn value (EV)*: the budget associated with the authorized work that has been completed.
- *Actual cost (AC)*: total cost incurred in accomplishing the work that the EV measured.
- The *Schedule variance (SV)* and the *Schedule performance index (SPI)* are used to assess the magnitude of schedule variations
 - $SV = EV - PV$
 - $SPI = EV / PV$

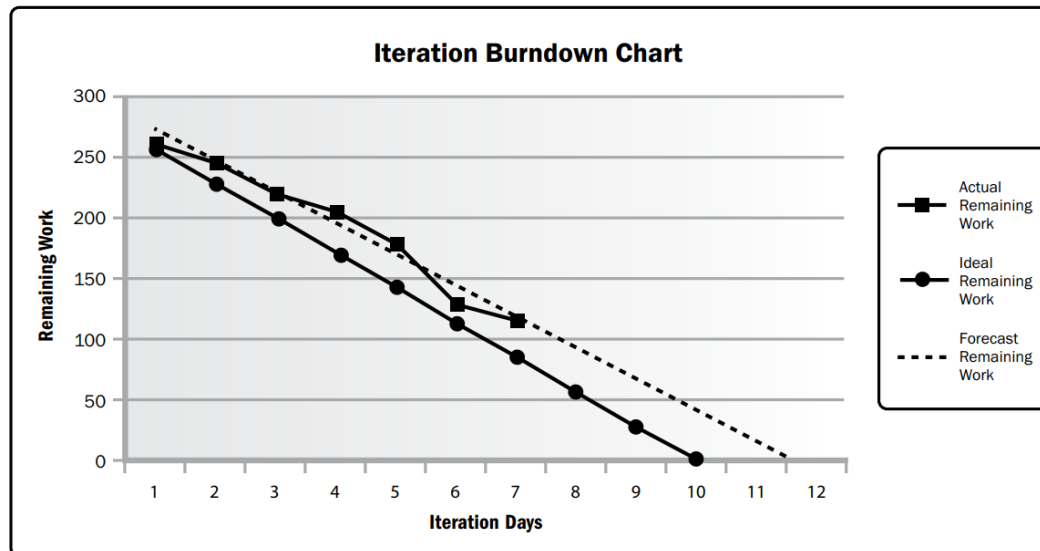


Tools & Techniques: Data analysis

Iteration burndown chart

Tracks the work that remains to be completed in the iteration backlog (Agile).

- Diagonal line representing the ideal burndown and daily actual remaining work is then plotted.
- Forecast trend line can be used to predict the likely variance at iteration completion.



Tools & Techniques: Data analysis

Performance Reviews

- Performance reviews measure and analyze schedule performance such as actual start and finish dates, % complete, and the remaining duration for the work in progress.

Variance analysis

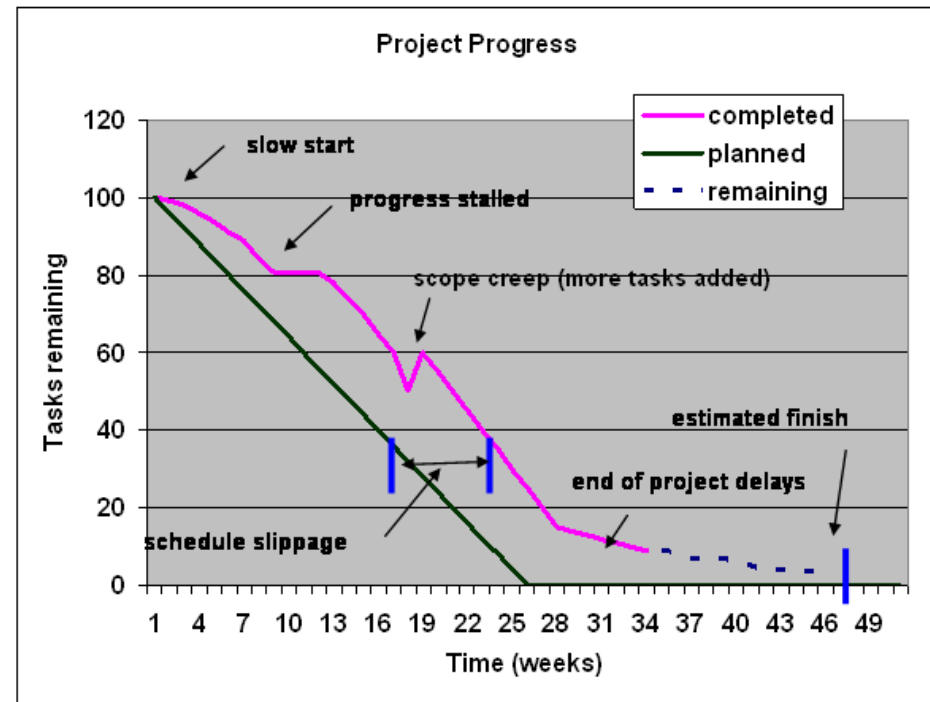
- Trend analysis examines project performance over time to determine whether performance is improving or deteriorating.

Trend analysis

- Variance analysis looks at variances in planned versus actual start and finish dates, planned versus actual durations, and variances in float.

What-if scenario analysis

Used to assess the various scenarios guided by the output from the Project Risk Management processes



Tools & Techniques

Critical path method

- Determine variance on critical path and near critical path
- Comparing the progress along the critical path can help determine schedule status.
- The variance on the critical path will have a direct impact on the project end date.
- Evaluating the progress of activities on near critical paths can identify schedule risk.

Leads and lags

- Is applied during network analysis to find ways to bring project activities that are behind into alignment with the plan.

Resource optimization

- Resource optimization techniques involve the scheduling of activities and the resources required by those activities while taking into consideration both the resource availability and the project time.

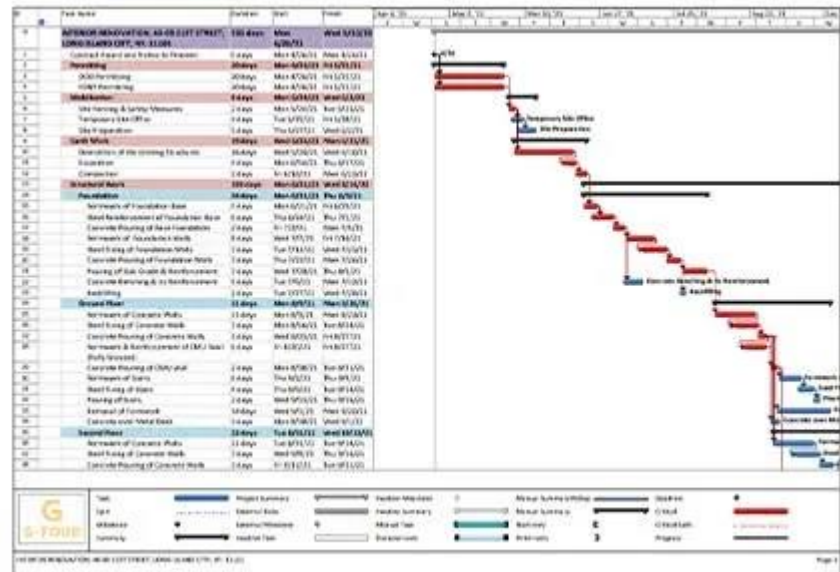
Schedule compression

- Are used to find ways to bring project activities that are behind into alignment with the plan by fast tracking or crashing the schedule for the remaining work.

Tools & Techniques

Project management information system (PMIS)

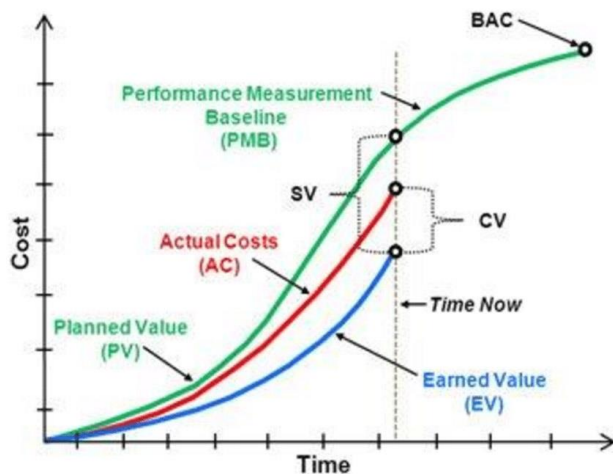
- Scheduling software that provides the ability to track planned dates versus actual dates, to report variances to and progress made against the schedule baseline, and to forecast the effects of changes to the project schedule model.



Outputs

Work Performance Information

- Information on how the project work is performing compared to the schedule baseline
- For project using EVA:
 - Schedule variance (SV)
 - Schedule performance index (SPI)



Schedule Forecasts

- Prediction of future performance based on available information and knowledge.
- The information is based on the project's past performance and expected future performance based on corrective or preventive actions.
- May include Earned Value performance indicators (SV, SPI).

Outputs

Change Requests

- Preventive actions may include recommended changes to eliminate or reduce the probability of negative schedule variances.
- Approved schedule baselines shall be only updated through integrative change control.



Project Management Plan Update

- Schedule Management Plan
- Schedule baseline
- Cost baseline

Project Document Updates

- Schedule data
- Project schedule
- Risk register

Inputs

Work performance data

- Activities status: remaining duration, percent complete...

Project management plan

- Scope baseline
- Schedule baseline
- Schedule management plan
- Performance measurement baseline.

Project documents

- Project schedule
- Project calendar: allow different work periods for some activities
- Schedule data
- Resource calendar
- Lessons learned register.

OPAs

- Existing formal and informal schedule control-related policies, procedures, and guidelines;
- Schedule control tools; and
- Monitoring and reporting methods to be used.